

A TECHNOSIGNATURE SEARCH OF ARCHIVAL ALMA OBSERVATIONS TOWARD 88 STARS IN 81 SYSTEMS WITHIN 40 PC

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ABSTRACT

We searched a defined subset of public ALMA observations covering 88 stars in 81 systems within 40 pc for unresolved spectral emission consistent with an artificial carrier: 443 spectral windows spanning 113.9 GHz of gross frequency coverage between 89.6 and 873.1 GHz, 109.4 GHz of it unmasked search space. Only 126 windows, toward 59 systems, are channelised finely enough for a drift-resolved carrier search; the other 317 constrain persistent power within a 15.6–31.25 MHz channel. The limits apply to signals confined within one native channel, far wider than a dedicated narrowband search uses. The deepest window-specific threshold, corrected for spectral response and intra-integration smearing, is 3.7×10^{13} W effective isotropic radiated power (EIRP); measured recovery at that trigger is 42–61 per cent by configuration and 89 per cent at 6σ . 20 windows cross the trigger and four are spatial outliers: three are circumstellar or foreground CO, two toward β Pictoris and one toward HD 48370; the fourth, toward CP–72 2713, is an unconfirmed single-epoch event with no astrophysical attribution, persistent emission of that strength being excluded on a 2.05-h baseline. **We identify no persistent, independently confirmed technosignature; one unattributed single-epoch event did not recur 2.05 h later.** A calibration sample of 603 windows shows the control statistic to depend on radius from the pointing centre, so the spatial ranking is treated as an empirically calibrated screen, not an exact rank test. The sample is work-list complete but archive-incomplete and over-represents F and G stars, 21 classified M dwarfs of 5 908 carrying a searched window, so it constrains ALMA’s observed stars, not the population. The survey’s value is to establish what sensitivity, completeness and false-positive behaviour heterogeneous ALMA archival data can support.

Subject headings: extraterrestrial intelligence, radio lines: stars, submillimetre: stars, surveys, astrobiology

1. INTRODUCTION

A technological civilisation, signalling deliberately or going about its affairs, may imprint the electromagnetic spectrum detectably, a *technosignature* (Sheikh 2020; Vidal et al. 2026). Six decades of radio searches cover many thousands of stars yet sample only 6×10^{-18} of the haystack of transmitter frequency, bandwidth, power and duty cycle (Wright et al. 2018). Three gaps persist. The millimetre and submillimetre regime is very sparsely searched, and everything above that axis’s ~ 115 GHz upper limit more sparsely still. Most surveys are aimed at interest-selected classes of star, leaving the local stellar population unsampled as a population. Few reprocess *existing* interferometric archives.

This paper addresses all three, asking how strongly existing ALMA observations constrain unresolved artificial spectral emission from nearby stellar systems. The sample it uses characterises the stars ALMA has observed, over-representing F and G stars and under-representing M dwarfs (§3), and every limit inherits that bias. We mine the archive, correcting for proper motion and Doppler drift, for unresolved spectral carriers (channel-confined excess at native ALMA resolution) and anomalous millimetre continuum. Throughout, “narrowband” describes the assumed intrinsically narrow trans-

mitter and never the resolving power of the data (§4). Channelisation sets which of two search classes a window belongs to, and the completeness of each class is measured in §4.5. The distinction is not a detail: 72 per cent of the searched windows have channels of 15.6 MHz or wider and carry no drift discrimination at all, so they constrain persistent channel-confined excess power and not a recognisable drifting carrier. Only 126 windows, toward 59 of the 81 systems, support the fine-channel search that “narrowband SETI” normally means. Three contributions follow. The first is the systematic exploitation of archival ALMA stellar fields, 109 target/band datasets mined from public data with no new observing time. The second is spatial-control screen calibration for interferometric SETI, in which the same statistic is evaluated at the star and at 512 control positions so that the false-alarm rate is measured and not assumed. The third is a quantitative accounting of the frequency, power and drift domain actually searched, which gives the non-detection explicit boundaries. No ancillary analysis carries a technosignature disposition. One worked case runs through the paper, a crossing towards β Pictoris, which is the largest star-versus-control contrast here and is attributed to the disc’s known circumstellar CO across two transitions and three epochs (§5.3). Figure 2 is the roadmap: it gives the selection from catalogue to searched sample on the left, and the decision chain from window to disposition on the right.

Two disclosures are made here rather than in an appendix. First, the statistic and the molecular-line mask

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were both defined on these data, so the rank probabilities and false-alarm figures describe this dataset, do not independently validate the pipeline’s false-positive rate, and are exploratory rather than pre-registered in any conventional sense; what we claim is narrower, that the criteria were repository-timestamped before the validation sample was analysed. An out-of-sample check on 603 windows searched after both were frozen is in §5.3.1. Second, the primary statistic was revised once during the analysis, from a maximum over a 135-position region around the star to the symmetric single-position statistic of Eq. 2, and the revision postdates the first flag it removes (§5.4). It is not circular, for a reason independent of any window: comparing a region maximum with single-position controls gives the star a rank floor of $135/(135 + 512) = 21$ per cent rather than $1/(512 + 1)$, so the old statistic flags at its own floor. Over the whole survey it flags 27 windows, 23 per cent of its 119 crossings, against 4 under the symmetric statistic: the revision removes a systematic inflation affecting 7 times as many windows as the one candidate whose disposition changed. Both statistics are carried for every window in the released catalogue (columns `star_snr_regionmax` and `stage1_flag_regionmax`), and Table 9 audits every window the original flagged.

2. BACKGROUND

Radio searches, from Cocconi & Morrison (1959) and Drake (1960) through the programmes reviewed by Tarter (2001) and Project Phoenix (Backus & Project Phoenix Team 2002), divide into wide-field commensal and targeted surveys (Zhang et al. 2020; Czech et al. 2021; Morrison et al. 2023; Li et al. 2026; Johnson et al. 2023), with wide-field low-frequency coverage from the MWA (Tremblay et al. 2024) and a single-dish L/C-band programme at the Sardinia Radio Telescope (Manunza et al. 2025), with machine-learning interference rejection now central (Ma et al. 2023; Choza et al. 2024; Jacobson-Bell et al. 2025; Poznanski et al. 2025; Garrett et al. 2026), a capability our classical threshold search lacks. Essentially all of that work lies below ~ 10 GHz, and the millimetre regime above it has been searched only once before the present work, with ALMA (Mason et al. 2024). There are physical reasons to go higher: interstellar scattering and dispersion fall steeply with frequency, a given aperture delivers a tighter beam per watt of EIRP, and directional communication, radar and beamed power are already engineered as coherent mm-wave emission. Atmospheric transmission, receiver noise and phase stabilisation raise the cost without forbidding it.

Mason et al. (2024) searched archival Band 3 windows toward 28 “stellar bycatch” stars, observed for other science and usable for free, placing $\text{EIRP}_{\min} > 7 \times 10^{17}$ W for the nearest and identifying the challenges of high-frequency work: drift rates scale with frequency, smearing erodes sensitivity without a drift search, and the dense forest of molecular lines raises the confusion background. Our pilot (White 2026) extended that methodology to TRAPPIST-1 across Bands 3, 6 and 7 ($\text{EIRP}_{\min} \sim 5\text{--}10 \times 10^{13}$ W, no detection) and built the ranked 100-star list this survey grew from. Every window here was re-reduced and re-searched from the raw archive data with the pipeline of §4, so where a star appears in both, TRAPPIST-1 included, the numbers to cite are this pa-

per’s frozen products.

Figure 1 places the nominal EIRP thresholds among published searches in *total power*; vertical position carries no technosignature merit. This survey’s competitive EIRP toward the nearest systems comes predominantly from the d^2 factor, and only weakly from sensitivity to Hz-wide carriers, to which a 15.625-MHz channel and a 2.8-Hz SETI channel respond very differently (§4). Placed against that literature, this survey extends the searched frequency range by nearly an order of magnitude and reaches EIRP thresholds comparable with cm-wave archival work for the nearest systems, while sampling far fewer stars, far fewer epochs and a channelisation seven orders of magnitude coarser than a dedicated narrowband search; it is complementary to those surveys and not a deeper version of them.

A non-detection must become a quantitative exclusion statement: Wright et al. (2018) formalised the “Cosmic Haystack” search volume, Margot et al. (2023) its conversion into an upper bound on the transmitter-hosting fraction, and Sheikh et al. (2025) asked where Earth’s technosignatures would be detectable. That bound is valid only if the sample represents the underlying population, which interest-selected samples do not. Most surveys select by proximity, spectral type, planets or Earth Transit Zone membership; the ALMA-specific form, disc selection, skews the sample young and dynamically active. We instead take every star within 40 pc lying in the usable field of at least one public ALMA observation (§3), a sample *complete with respect to the frozen candidate work list but archive-incomplete*: of 168 catalogue crossmatches, 115 correspond to genuine ALMA coverage and 88 were searched, and only a fraction of the execution blocks those stars hold entered the analysis. It makes no completeness claim for the local population itself. That choice removes technosignature-specific selection and leaves the archival astrophysical selection in place; the 168 sample entries are ~ 1 per cent of the catalogued population within 40 pc (the 88 actually searched here are ~ 0.5 per cent), and the residual coverage selection is characterised as a completeness function (§3, Table 5).

3. SAMPLE SELECTION

The sample is *archive-target-complete* but epoch-incomplete within 40 pc, a much weaker condition than volume completeness and defined with its counterpart in Table 1. The procedure admits every star within 40 pc of the Sun (Gaia DR3 parallaxes, distances direct inversions $d = 1/\pi$; every entry $\pi \geq 25$ mas with a parallax measured to better than 20 per cent, which is the condition the archive query applied) whose proper-motion-corrected position at the observation epoch falls in the quoted field of view of at least one public ALMA execution block. The star need not sit at the pointing centre, and §5.2 shows the quoted field of view admitted 53 stars no observation in fact contains, so “covered” is a property of our selection and not a guarantee. Disc, planet and spectral type enter as descriptive metadata only. Units are defined in §4.1. This gives 168 unique target stars, processed nearest-first. Because trigger thresholds scale with d^2 at fixed noise, that order is sensitivity-ranked, so the prevalence statements of §6 are conditional on it.

Composition follows from the archival selection and is

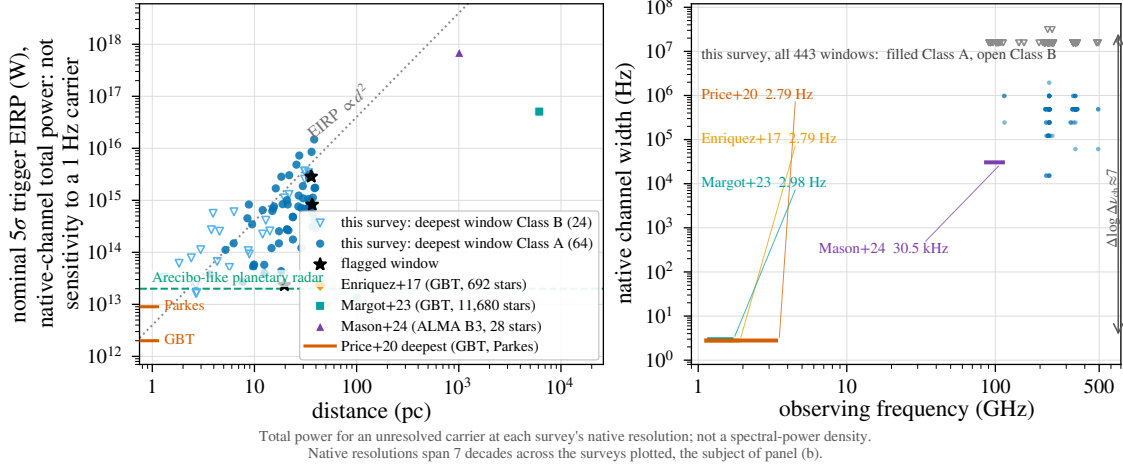


FIG. 1.— Literature context of the 5σ trigger thresholds, in total power (§2). (a) EIRP against distance. Blue: this survey, deepest window per star (88 stars, 1.30–39.63 pc; filled: Class A, fine-channel; open: Class B, coarse-channel); orange: Enriquez et al. (2017); aqua: Margot et al. (2023); violet: Mason et al. (2024); left-edge ticks: deepest per-target limits of Price et al. (2020). Grey dotted: $\text{EIRP} \propto d^2$ at constant received flux. Green dashed: the Arecibo 2.38-GHz EIRP benchmark, a power scale only (§4.3). (b) Native channel width against observing frequency for the same windows and programmes: resolutions differ by $\sim 10^7$, so the panels must be read together and a lower EIRP here does not imply greater sensitivity to a hertz-wide carrier. Plotted values are nominal 5σ triggers: they exclude both the $\times 2.29$ median spectral-response correction and the $\times 1.41$ available from the individual parallel hands, so a transmitter comparison carries a further $\times 3.2$ against this survey’s curve at the median response, more in the tail.

TABLE 1

NOMENCLATURE FOR SEARCH OUTPUT AND THE TWO SENSES IN WHICH A PROBABILITY IS QUOTED.

Term	Meaning
crossing	one channel $\times$ drift cell at the stellar position with $T \geq 5$: a threshold crossing (state 1); counted per window in the main text and per cell in Appendix F
stage-1 spatial outlier	a window whose T_* exceeds all 512 control statistics (“star-exceeds-ring”), a screening device that carries no candidate significance (state 2); shortened to “stage-1 outlier” after first use
$P_{\text{rank}, \text{min}}$	$1/(N_{\text{ctrl}} + 1) = 1/513$, the rank <i>resolution</i> of the ensemble and not an attainable false-alarm probability, the controls being only approximately exchangeable with the star (§5.3.2); the rank <i>resolution</i> of one window’s control ensemble: the smallest per-window probability the design can express
$P(\text{false trigger})$	the survey-level false-positive probability after thresholding, correlation and non-Gaussianity. The two differ, the first being a property of one window’s ensemble and the second of the whole survey, and the survey-wide Bonferroni scale 1.2×10^{-4} lies a factor 16 below the first (§5.3)
work-list complete	every star on the frozen candidate work list of §3 that proves genuinely covered is searched. We avoid “target-complete” and “archive-complete”, which invite the reading that every ALMA-observed star within 40 pc, or all available data, are searched; neither is the case
epoch-incomplete	only 104 of the 451 public execution blocks those targets hold are searched

tabulated against the reference census in Table 10. The sample is F- and G-enriched and deficient in M dwarfs, and 39 of the 443 windows come from 3 systems; §6.3 gives the figures and their cost. The spectral-class proxy is `teff_gspphot`, available for 52 of 88 sample stars and missing preferentially for the coolest objects, so the M fraction is a lower limit and the earlier-type ratios upper limits. The debris-disc hosts contribute dusty continua,

kept from masquerading as anomalies by the continuum lane’s screen (§4).

The 168 stars are complete against, and conditional only on, the 40-parsec reference catalogue intersected with the public-archive snapshot of 2026 September 8. Against that Gaia DR3 catalogue of 17 566 stars they are $\lesssim 1$ per cent and the 88 searched are 0.50 per cent, fewer still once §5.2 removes those never observed (Fig. 2). The two parallax cuts are not interchangeable, the sample’s binding condition being $\sigma_{\varpi}/\varpi < 20$ per cent against the census’s 10 per cent, and the realised sample is better than its condition, reaching 2.48 per cent at worst. Because the census is incomplete at both ends, the fraction shows only that coverage is small.

Stellar counts throughout are catalogue-entry counts. The 88 searched entries comprise 81 independent systems, differing by seven designation-linked component pairs (SIMBAD designations; per-entry rows, distances and unresolved-pair notes in the machine-readable catalogue). The 109 star/band datasets, 443 windows and 104 execution blocks therefore derive from 81 systems. The 443 rows comprise 408 distinct (execution block, spectral window) datasets, and the larger number is used as the trials denominator throughout, which is conservative.

“Usable public ALMA coverage” is a five-criterion decision tree, frozen absent an erratum: (i) at least one public execution block (EB) whose visibilities are the archive’s delivered QA2 products, taken as delivered and not recalibrated, so calibration vintage varies across the 2013–2025 span; the flux-scale term this contributes is small beside the $\times 2.29$ spectral-response correction. They have passed ALMA’s second-stage quality assurance (QA2) and cover the star’s epoch-propagated position, with no minimum integration, sensitivity, antenna count, mosaic-tile count or array configuration beyond QA2; (ii) that position inside the field’s primary beam, the extraction aborting beyond $2\theta_{\text{PB}}$ and the largest retained offset being $0.46\theta_{\text{PB}}$, so nothing approaches the bound; (iii) each

spectral window admitted individually, on QA2 vetting alone; (iv) no programme-category restriction, science, calibration and observatory projects entering on identical terms; and (v) a fixed archive snapshot date (2026 September 8). No star or window was excluded by an unlisted criterion, and none for short integration.

Completeness with respect to the work list holds in targets and tunings, not in epochs or integration time. The 102 member observing unit sets that supply the searched blocks hold 451 public execution blocks between them, of which 104 were analysed (23.1 per cent): the survey takes at most 4 blocks per target and de-duplicates spectral windows by identifier across the blocks of one set. 75 sets therefore hold at least one unanalysed block, supplying 304 of the 443 windows and 62 of the 88 stars. Every epoch statement here concerns analysed epochs, and the archive would support a stronger one. A further 150 blocks processed under the same frozen pipeline are used in §5.3.1 as an external calibration sample and enter no count in this section.

ALMA project metadata quantify pointed versus serendipitous coverage: almost every spectral-window-covered star is the pointed science target in at least one execution block, leaving Wolf 219 the sole serendipitous star, so the residual selection function of a work-list-complete sample is the union of ALMA target lists.

4. DATA AND METHODOLOGY

The design is one shared archive-ingestion stage feeding two independent search lanes, and Table 3 gives the data path of the primary lane in order. For each target we mine, within a bounded budget, the calibrated archival execution blocks covering the star’s Gaia-propagated position. Their visibilities feed (i) the *primary* lane, a Doppler-drift-corrected spectral-carrier search following Mason et al. (2024) and Enriquez et al. (2017), which converts each non-detection into an equivalent isotropic radiated power limit $\text{EIRP}_{5\sigma} = 4\pi d^2 S_{\min} \Delta\nu$. The product $S_{\min} \Delta\nu$ is an integrated power only if the channel’s effective noise bandwidth equals its separation, which for a smoothed correlator it does not; that is precisely what the response factor C_{response} corrects, and the correction is *calibrated rather than assumed*. Injecting unresolved tones of known amplitude into real visibilities and recovering them through the unmodified pipeline returns $\times 2.02$ – $\times 2.69$ (median $\times 2.31$) across correlator modes, against the analytic $\times 2.29$ expected for the Hanning response (Appendix B), so $P_{\text{eff}} = S_{\min} \Delta\nu C_{\text{response}} C_{\text{smear}}$ recovers the injected power: the medians agree to one per cent, the quoted range being window-to-window variation in the correlator response and not an error on the estimator. EIRPs are quoted to two significant figures, which is the precision the flux scale supports. Here $S_{\min}$ is the primary-beam-corrected 5σ flux density defined below, $\Delta\nu$ the native channel width and d the distance (worked numerically in Appendix A). Five terms enter the systematic budget, each multiplying $S_{\min}$ and hence $\text{EIRP}_{5\sigma}$ directly: the ALMA absolute flux scale, 5–10 per cent and band- and epoch-dependent; residual atmospheric phase after water-vapour correction and phase referencing; the primary-beam model at the stellar offset; visibility calibration; and the instrumental spectral response, median $\times 2.29$, the largest of the five and carried explicitly as

P_{eff} by the reading rule below. *Gaia* distance errors are negligible. The atmospheric term is a coherent point-source loss $\exp(-\sigma_\phi^2/2)$, 5–20 per cent at 230–345 GHz, common to star and annulus within the isoplanatic patch and so no threat to exchangeability. The same visibilities feed (ii) an *ancillary* line-excluded continuum lane (Appendix E). In most windows an Hz-wide carrier is unresolved within a 15.625 MHz channel, so the data cannot show a one-channel excess to be intrinsically narrow. Two survey classes are reported, distinguished by the dimensionless drift parameter

$$\eta_{\text{drift}} = |\dot{\nu}_{\text{max}}| \tau_{\text{track}} / \Delta\nu_{\text{ch}}, \quad (1)$$

the channel-width fraction the *maximum* searched drift $|\dot{\nu}_{\text{max}}|$ traverses over the full on-source track τ_{track} at native channel width $\Delta\nu_{\text{ch}}$. The track duration is written τ_{track} throughout, leaving T for the search statistic alone. The track is the right time-scale, the de-drift step summing along it; η_{smear} (Appendix A) is the same ratio over one integration. A window is *drift-resolving* when $\eta_{\text{drift}} \gtrsim 1$, carrying a carrier across enough channels for its motion to be measured. The drift ceiling inside η_{drift} is a search-grid boundary, the widest trial drift the stack evaluates, not a prior on transmitter drift rates; carriers drifting faster are missed, and §6.1 gives the resulting selection function (Fig. 8).

Class A is the fine-channel spectral-carrier search: the 126 windows of fine channelisation, native widths 15.259–1953.125 kHz (median 488.3 kHz; 62 per cent at the 488.3-kHz mode), spanning $\eta_{\text{drift}} = 0.34$ –422 (median 12.8). It is the only class with measured end-to-end recovery of a drifting carrier. *Class B* is the coarse-channel spectral-excess search: the 317 coarse windows, 72 per cent of the sample, native widths 15.625–31.25 MHz (309 windows at the 15.625-MHz mode), spanning $\eta_{\text{drift}} = 0.004$ –1.35 (median 0.36). It is sensitive to persistent near-stationary excess power, carries no drift discrimination, and is never described as a narrowband search anywhere in this paper.

η_{drift} is the physical criterion and the classes are labelled by the instrument. Channelisation is bimodal, with no window between 1953.125 kHz and 15.625 MHz, an empty gap of factor 8.0, so any boundary inside the gap assigns identical membership and figures label the classes by name. The two criteria do not coincide exactly, and the overlap is quantified: 12 long-integration Class B windows reach $\eta_{\text{drift}} \geq 1$, and only marginally (maximum 1.35), where a one-to-two channel traverse supports no discriminative drift fit after baseline removal; 6 short Class A windows fall below unity and are conservatively kept in Class A. On the physical criterion alone 132 windows of 443 reach $\eta_{\text{drift}} \geq 1$, against 126 in Class A, so the labels are channelisation labels and “drift-resolved” is used of the class only where the two agree. Each window’s η_{drift} , η_{smear} , trial drift count and maximum searched acceleration $a_{\text{max}} = 3.6$ – 4.0 m s^{-2} are columns of the released catalogue, so any reader may reclassify on the physical criterion alone. Results are given per class first and combined only where statistically appropriate (§5.3).

Both lanes exclude catalogued molecular transitions at tolerances suited to their statistics. The continuum lane masks the common species at native resolution be-

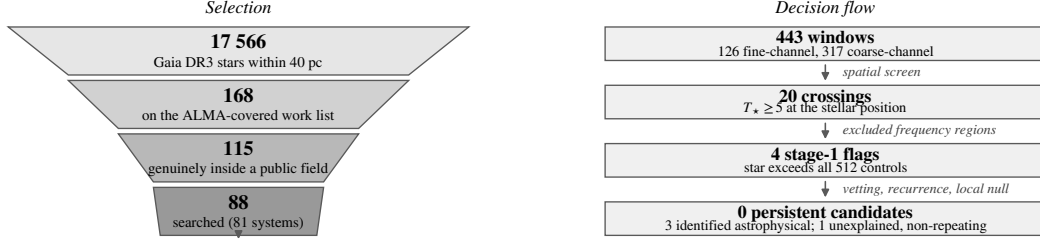


FIG. 2.— Left: the selection funnel. The three denominators the text keeps apart are the catalogued population, the 168-entry ALMA-covered work list, the ~ 115 stars a public field genuinely contains, and the 88 searched. Right: the decision flow from spectral window to disposition. The 512-control comparison is a candidate-generation screen, and the gates in *italic* are where a crossing can leave the chain.

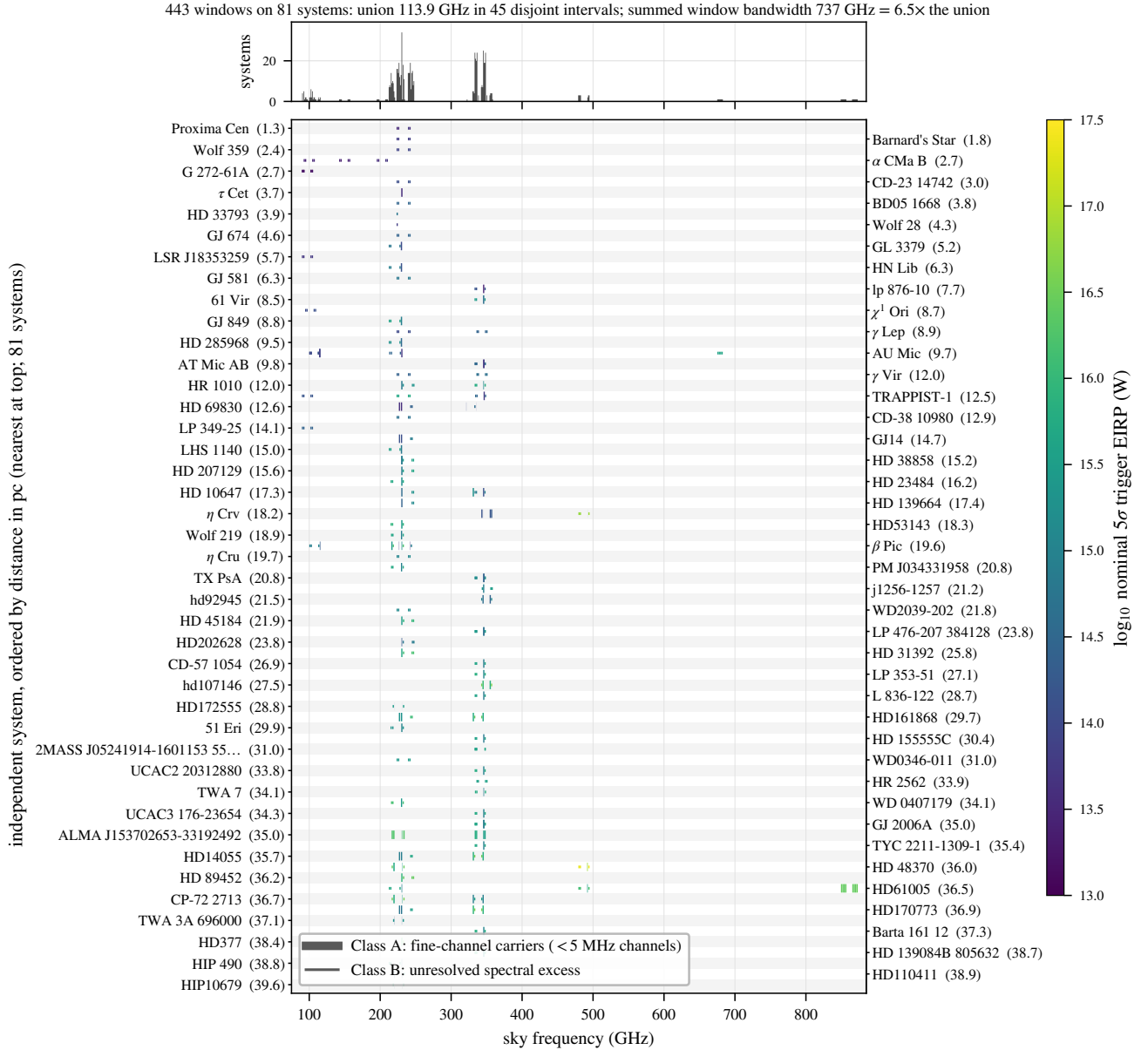


FIG. 3.— What this survey sampled in frequency. Each searched window is a horizontal segment on its host system's row, rows ordered by distance and labelled alternately on the left and right axes, coloured by nominal $\text{EIRP}_{5\sigma}$, with line weight distinguishing the fine-channel (Class A) from the unresolved-spectral-excess (Class B) channelisation; the upper panel counts systems per gigahertz of sky frequency. The coverage is narrow, clustered and repeatedly revisited: 45 islands concentrated near 230 and 345 GHz, with summed window bandwidth $6.5\times$ the union. The union is the frequency coverage; the summed bandwidth is the integrated search effort, and the two answer different questions (Scope of the constraints, §4).

TABLE 2

THE SENSITIVITY AND COMPLETENESS QUANTITIES OF THIS PAPER, AND WHAT EACH IS FOR. THE ABSTRACT AND CONCLUSIONS QUOTE P_{eff} ; THE CATALOGUE CARRIES ALL OF THEM PER WINDOW.

Quantity	Meaning and use
P_{trig}	nominal 5σ channel threshold; the pipeline’s trigger, an internal statistic
C_{resp}	spectral-response correction, $\times 2.29$ median, window-specific
C_{smear}	intra-integration drift smearing, $1/\eta_{\text{smear}}$
P_{eff}	$P_{\text{trig}} C_{\text{resp}} C_{\text{smear}}$: the power to compare with a transmitter EIRP. Quote this one
P_{50}, P_{90}	injected amplitudes recovered half and nine-tenths of the time; completeness, not threshold
Class A/B	fine-channel / coarse-channel windows; different experiments, never pooled in a completeness statement
dwell	recovery against the fraction of the track a signal is present

fore integrating; the carrier lane applies a $\pm 50 \text{ km s}^{-1}$ stellar-frame velocity exclusion mask (§5.3.10) to any threshold crossing. That half-width was widened after the β Pictoris crossings had been seen, so the safeguard belongs here rather than in a later section: reclassifying all 20 crossings at $\pm 20, \pm 30, \pm 50$ and $\pm 100 \text{ km s}^{-1}$ admits no new unattributed outlier, and at every width CP–72 2713 remains the only unmasked one. The mask is *excluded search space*: the unmasked survey covers 113.9 GHz of unique sky frequency, the effective default quoted throughout is the $\approx 109.4 \text{ GHz}$ surviving it, and no constraint is placed on carriers inside the excluded velocity regions. Masking reduces the frequency space a candidate can occupy while leaving the per-channel noise untouched, so no threshold moves. All flux densities and EIRP thresholds are total intensity (Stokes I), summed over the two parallel-hand cross-correlations with QA2 weights; no cross-hand extraction or circularity test was performed, so the construction is polarisation-blind in amplitude and a strongly polarised artificial signal responds like an unpolarised one at fixed total power. The estimator is safe in this respect, though the data have not been audited for it. Both parallel hands are delivered for 104 of the 104 searched execution blocks, which is all of them, so the discrimination described next is available from the existing data and needs no new observation. What the retained products do not record is per-correlation flagging inside QA2, so we cannot say what fraction kept both hands unflagged all the way through; one that lost a hand would be $\sqrt{2}$ less sensitive than its threshold implies, with nothing here to catch it. What Stokes I costs is discrimination, not sensitivity: an artificial coherent carrier is more likely to be strongly polarised than the unpolarised line and dust emission that fills this archive, so a polarisation measurement would separate the two for free.

How to read every limit in this paper. Table 2 collects the quantities and says which is which; three of them are kept apart throughout, and only the first is what the pipeline acts on.

The **trigger power** P_{trig} is the nominal EIRP $_{5\sigma}$: five times the searched window’s per-channel rms, at that window’s native channel width, for a transmitter at the

star’s position matching the searched morphology. It is the level at which the search fires, it is what every “ 5σ ” below means, and it is what the figures label as the nominal trigger.

The **effective unresolved-carrier threshold** P_{eff} is P_{trig} corrected for the instrument’s spectral response. ALMA applies online Hanning smoothing, which leaves only 0.38 to 0.50 of a sub-channel tone’s power in the peak channel depending on where in the channel it falls, median 0.44 (Appendix A). Nominal thresholds are therefore optimistic by $\times 2.00$ to $\times 2.67$, median $\times 2.29$, and $P_{\text{eff}} = 2.29 P_{\text{trig}}$ is the number to compare with any transmitter power. It follows from the 0.25/0.5/0.25 kernel and is window-specific: $\times 2.29$ where the archive reports an effective resolution of 2.00 channel spacings (403 windows) and $\times 1.33$ –2.00 for the 28 with online channel averaging. Injecting through the response kernel returns $\times 2.02$ at channel centre and $\times 2.69$ at the edge, median $\times 2.31$, within 1.1 per cent of the analytic value (Appendix B). It is a sensitivity correction and not a completeness, it is not small, and we treat it as the principal physical threshold.

The **completeness limit** P_x , the power at which measured recovery reaches x per cent, is a different quantity again: $P_{50} = 1.10 P_{\text{trig}}$ and $P_{90} > 2 P_{\text{trig}}$ on the calibration window’s drifting class (Appendix B). Transfer to other bands, channelisations and integration times is no longer assumed but measured on 28 further configurations, where P_{50} spans 0.64 – $1.13\times$ its median (Appendix B); P_{95} remains *not determined*. What this survey has is a measured reference recovery function whose transferability is now bounded by measurement, and not a measured Class A completeness function.

Every headline EIRP is quoted beside its native channel width, and the catalogue carries P_{trig} and both response-corrected values per window.

Three signal morphologies are detectable here, and §6.1 quantifies each against real windows: a continuous frequency-stationary carrier, a continuously drifting one, and an intermittent one.

4.1. Statistical units and independence

Six units appear below and they are not interchangeable, so we fix them here and say which one carries which probability. A *catalogue entry* is one star as SIMBAD designates it, 88 of them searched; a *physical system* is the set of entries gravitationally bound together, 81 of them, seven designation-linked pairs accounting for the difference. A *target/band* dataset is one entry in one ALMA band, 109 in all. An *execution block* is one archival observation, 104 searched, and it is the unit that shares weather, calibration and correlator setup. A *spectral window* is one window of one execution block: 443 catalogue rows, which reduce to 408 distinct (execution block, window) datasets once repeated archive products are collapsed. An *independent frequency trial* is neither of those: within a window the channel $\times$ drift grid is oversampled on both axes, so its 177 – 6866 208 cells carry about a factor 5.3 fewer independent trials than cells (Appendix F). An *independent epoch* is a separate execution block at the same tuning, and blocks of one scheduling block executed on one night are not independent epochs in any useful sense.

Which unit enters a population-level statement therefore depends on the statement. Per-window rank probabilities are conditional on the window and use no denominator at all. The survey-level flag expectation uses 443 windows, which is conservative because the 408 distinct datasets are the smaller number and because windows within a block are correlated; 443 windows are not 443 independent trials, and no probability here should be read as though they were. Anything about how common transmitters are must use 81 systems, never 88 entries and never 443 windows. Anything about persistence uses epochs, of which most stars here have one.

4.2. The search statistic, and the words used for its output

The primary statistic is formed on extracted spectra, never on images or visibilities. It operates on the QA2-calibrated measurement set as delivered, per integration and per native channel. The archive’s flagging carries through untouched and no spectral regridding or interpolation is applied (Table 3, step 3). What the extraction returns at a propagated sky position $\mathbf{x}$ is the real part of the weighted vector average of the phase-shifted visibilities, which is the natural-weighted dirty-map value there. There is no deconvolution, so a quoted S_{peak} in a field with extended emission is a dirty-map amplitude. Baseline removal follows extraction (step 4). The primary-beam correction enters the quoted thresholds and fluxes (§4.3) and leaves untouched the extracted amplitudes the statistic ranks. For a sky position $\mathbf{x}$, each per-integration spectrum $I(\mathbf{x}, t, \nu)$ is baseline-subtracted by a block median along the *frequency* axis. No temporal median is applied at any stage (§4.4). The residual dynamic spectrum is stacked along each trial linear drift rate $\dot{\nu}$ with inverse-variance weights, and the statistic is the maximum of that stack over the channel $\times$ drift grid,

$$T(\mathbf{x}) = \max_{\dot{\nu}, \nu} \frac{\sum_t I(\mathbf{x}, t, \nu + \dot{\nu}t) / \sigma^2(t, \nu)}{[\sum_t 1 / \sigma^2(t, \nu)]^{1/2}}, \quad (2)$$

a matched filter whose denominator is the noise of the stacked spectrum. Only when σ is the same for every integration and channel does it reduce to a single-channel amplitude divided by one noise scale; otherwise the weights vary along the track. The scale $\sigma(t, \nu) = 1.4826 \text{ MAD}_{i=1\dots 512} [I_i(t, \nu)]$, that is $1.4826\times$ the median absolute deviation taken *across the control positions* at each integration t and each channel ν , which makes it global in position, local in frequency and in time, and shared by the star and by every control.

The across-position median enters only in forming that deviation and is never subtracted from the numerator, so T is an absolute stacked signal-to-noise carrying a cross-position noise scale, and it measures no local contrast.

Why this form and not a region maximum. T is evaluated at one position, identically at the star and at each control, and that is what makes the rank interpretable. If instead the stellar side takes a maximum over n_{src} positions while the controls remain single positions, the two sides are no longer exchangeable: under the null the region maximum exceeds all N_{ctrl} controls with probability $n_{\text{src}}/(n_{\text{src}} + N_{\text{ctrl}})$, not $1/(N_{\text{ctrl}} + 1)$. With $n_{\text{src}} = 135$ and $N_{\text{ctrl}} = 512$ that is 21 per cent rather than 0.2, a floor under which most crossings flag whatever the data

contain. The argument is combinatorial and refers to no window: it holds for any statistic, any noise distribution and any field. An earlier version of this analysis used the region-max form and was revised for exactly this reason; §5.4 gives the chronology, including the fact that the revision postdates the first flag it removes, and Table 9 audits every window either form flags. A line filling the primary beam therefore drives star and annulus up together instead of cancelling.

The baseline filter takes the median in contiguous blocks of 65 channels and interpolates linearly between block centres, using $\max(2, n/65)$ blocks over n usable channels. Its effective width is therefore 65 channels in the 126 windows long enough to hold more than two blocks, and $n/2$ in the short coarse windows where the block count collapses to two. This sets a ceiling on the linewidth the filter passes: ≈ 32 MHz at the fine class’s modal channelisation, and ≈ 0.9 GHz at the coarse class’s (Appendix A). The crossing channel is *not* excluded from the scale. The median is taken across positions at a fixed channel, so what matters is how many of the 512 controls carry the feature. A compact feature at the star appears in none of them and the 50-per-cent breakdown point holds; emission filling the primary beam appears in all of them, which is the HD 48370 case (§5.3). $T_{\star} \equiv T(\mathbf{x}_{\star})$ at the proper-motion-propagated stellar position, and the same operation, drift grid, channelisation and noise map give $\{T_i\}$ at all 512 control positions.

The controls are placed by `probe_positions()` in the released extraction code, which adds its offsets to the star’s own direction cosines, so the annulus is centred on the *star* itself, wherever the star sits relative to the field phase centre. It holds 512 positions drawn uniform in area, $r = \sqrt{U(r_{\text{in}}^2, r_{\text{out}}^2)}$, and uniform in azimuth, from the fixed seed 20260825 re-applied per window. All windows therefore share one pattern in units of θ_{PB} , and two windows of one tuning sample identical sky positions (Fig. 5). Radii run $0.14\text{--}0.78\theta_{\text{PB}}$, with θ_{PB} evaluated at each window’s own median frequency, so the annulus scales window by window. A separate inner check-ring of 135 positions serves the point-source test, and the full per-window geometry is released as catalogue columns.

The frozen products carry none of that geometry, so it is reconstructed from the code, and two checks confirm the layout that ran: the inner check-ring it generates has the 135 positions the retired region-max statistic maximised over, and where extended CO fills the beam the control statistics fall with reconstructed radius, at Spearman $\rho = -0.49$ in the HD 48370 Band 6 window. Neither would survive a wrong position-to-index mapping.

Placing controls off-centre costs sensitivity. A star near the pointing centre sits at primary-beam response ≈ 1 while the annulus samples 0.95 at $0.14\theta_{\text{PB}}$ falling to 0.19 at $0.78\theta_{\text{PB}}$, area-weighted mean 0.47. Star and controls are therefore not exchangeable in flux density, and they remain exchangeable in the signal-to-noise ratio the statistic actually ranks, because $T(\mathbf{x})$ is formed on uncorrected amplitudes against one shared noise scale. Extended sky brightness does not cancel: smooth emission lives on short baselines whose phases barely change under a rotation of a few arcsec, so star and annulus rise *together*, as they do for HD 48370, while emission offset from the star raises particular controls above it, as

in the same star's ^{13}CO window (ring maximum 20.4 at 9.2 arcsec against 4.6 at the star). Two asymmetries remain and they run in opposite directions: a suppressed control can only make a star-exceeds-ring outcome more likely under the null, whereas a multiplicative calibration error scales with the continuum flux at the extracted position and so favours the star, where that flux is not zero.

Neighbouring controls are correlated below the synthesised-beam scale, so a window's annulus carries $N_{\text{eff}} \simeq 30\text{--}43$ independent spatial trials in the compact configurations where the symmetric reprocessing runs measure it. That figure is an analytic beam-geometry estimate, not a measured correlation length, and it is the worst case: the annulus contains far more resolution elements than the 512 positions drawn in all but those configurations (Appendix F). The rank test needs exchangeability rather than independence in any case, and correlation among controls costs resolution without touching the $1/513$ floor. The frozen products do not carry the synthesised beam or the array configuration, so radii are quoted in arcsec and in θ_{PB} ; the ALMA archive carries both, and what they say about this geometry, including the 32 per cent of windows taken with the ACA 7 m array against a θ_{PB} defined on 12 m, is in Appendix F.

Ranking T_* against 512 values fixes the smallest per-window p -value the design can express at $1/513 = 1.95 \times 10^{-3}$, while the survey-wide Bonferroni scale over 443 windows is $1.2 \times 10^{-4} = 0.05/443$, smaller by a factor of 16 (Appendix F). **No single event in this design can therefore reach survey-wide significance.** The 512-control comparison is a candidate-generation screen and never a significance test: a stage-1 spatial outlier selects a window for examination and establishes nothing about it. Table 1 fixes the vocabulary and Table 3 allows the statistic to be reimplemented from the main text alone.

Residual continuum after baseline removal, self-calibration residuals and position-correlated calibration error all concentrate at or near the phase centre, where the pointed science target sits, and σ is built from the annulus and carries none of it. These stars are not all blank there, the continuum lane detecting 17 of 57 target/bands (Appendix E). A multiplicative spectral calibration error carries the same asymmetry in the unfavourable direction, since a residual bandpass ripple of fractional amplitude ϵ leaves a channel-confined artefact of order ϵS_{cont} at the extracted position and essentially nothing on the annulus. The direct test of the premise is §5.3.5, which compares the star's own residual variance with the controls' in the four flagged windows and finds them equal to better than a per cent, and two indirect ones agree with it. Recomputing the scale locally, within a band around each feature and excluding the feature itself, gives local-to-global ratios of 0.97 (CP-72 2713), 1.29 (HD 48370), 0.93 (β Pic B3) and 1.02 (β Pic B6), which agree with the reproducible R_σ test above and carry the same conclusion: the global scale is accurate to a few per cent except towards HD 48370, where the field is filled with CO and rescaling both the statistic and its ensemble leaves the window flagged ($T_* = 21.0$ against 20.8), which is the disposition it already has. And because the same estimator runs at the star and at every control, a badly misstated σ would displace the con-

trol distribution, which the uniform pseudo-star ranks exclude. None of this reaches a per-crossing local scale inside the search, which remains an open item (§6.4).

A rank means something only when the identical operation runs at the star and at every control: under a pure-noise null a region maximum over n_{src} positions at the star exceeds all n_{ctrl} single controls with probability $n_{\text{src}}/(n_{\text{src}} + n_{\text{ctrl}}) = 20.9$ per cent for the 135-position check-ring, whatever the window and whatever the noise distribution. The released statistic therefore evaluates one position at the star and one at each control; §5.4 dates the earlier asymmetric version and audits the windows it affected.

The searched drift-rate range is the larger of (a) a generic frequency-scaled default ($12\text{--}13 \text{ Hz s}^{-1} \text{ GHz}^{-1}$), the Mason et al. (2024) fiducial with a threefold safety margin, and (b) for confirmed exoplanet hosts, the maximum line-of-sight orbital acceleration of the innermost known planet, so a transmitter on a short-period planet cannot drift out of the window. Drift rate maps onto line-of-sight acceleration through the first-order Doppler relation

$$a_{\text{los}} = c \frac{\dot{\nu}}{\nu}, \quad (3)$$

so the ceiling is quoted in cross-band-comparable form $\dot{\nu}/\nu$: $12\text{--}13 \text{ Hz s}^{-1} \text{ GHz}^{-1}$ is $(1.2\text{--}1.3) \times 10^{-8} \text{ s}^{-1}$, that is $|a_{\text{los}}| = 3.6\text{--}3.9 \text{ m s}^{-2}$ independent of observing frequency, the acceleration at ~ 0.04 au around a solar-type star. Earth-motion contributions lie two orders of magnitude below and are subsumed by the grid. The grid is uniform in $\dot{\nu}$ over n_{drift} steps (2–1944 across windows, recorded per window), so a carrier midway between trial rates leaves a residual drift traversing at most 0.26 of a native channel over the longest fine-class track, and the resulting amplitude loss is an order below the recovery slope measured between 5σ and 10σ . Two limits of the model matter more. The rates searched are *apparent* topocentric ones, so the survey does not separate a deliberately chirped carrier from an accelerating platform and constrains only linear apparent drift inside the ceiling; and an emitter need not be tied to a known planet at all, a free-flying platform or rotating structure accelerating as it chooses and falling outside these constraints without being excluded by them. For circular orbits the ceiling covers a transmitter co-orbiting an Earth analogue by three orders of magnitude and Proxima b comfortably, while TRAPPIST-1 b ($a_{\text{los}} = 4.00 \text{ m s}^{-2}$) sits at it and c (2.14 m s^{-2}) reaches it at modest eccentricity, which raises the periastron value by up to $(1+e)/(1-e)^2$ (Fig. 8). That value is the maximum over orbital phase, not a typical one. TRAPPIST-1 transits, so its inclination is measured and effectively edge-on, and the relevant figure is the edge-on one: TRAPPIST-1 b exceeds the ceiling over 5–29 per cent of its orbital phase, depending on which end of the $12.0\text{--}13.3 \text{ Hz s}^{-1} \text{ GHz}^{-1}$ ceiling applies, and is searchable through the rest of it.

Across the 443 windows maximum searched drift rates span $1.1\text{--}5.9 \text{ kHz s}^{-1}$ and trial drift rates per window 2–1944 (median 4), the largest being the narrow, finely channelised β Pictoris Band 6 window (§5.3.7). 7 windows are narrower than 100 MHz. The threefold margin matters. At $1\times$ the search would have been blind to two of its threshold-crossing windows, including the AU Mic

TABLE 3

THE STATISTICAL DATA PATH, IN ORDER. STEPS 1–8 RUN IDENTICALLY AT THE STELLAR POSITION AND ALL 512 CONTROLS; ONLY STEP 9 ONWARD DISTINGUISHES THEM.

1	calibrated measurement set from the ALMA archive
2	stellar position propagated to the observation epoch (Gaia DR3)
3	spectral extraction at that position, native channelisation
4	per-integration baseline removal: block median along frequency ($W = 65$)
5	stack along each trial linear drift rate
6	robust noise map $\sigma(t, \nu)$: MAD across the 512 controls
7	$T(\mathbf{x})$ from Eq. 2, an inverse-variance-weighted stack
8	repeat 3–7 at 512 control positions on a ring
9	threshold: $T_* \geq 5$ (a crossing)
10	rank test: $T_* > \max_i T_i$ (stage-1 spatial outlier)
11	molecular-line mask, $\pm 50 \text{ km s}^{-1}$ stellar frame
12	vetting: recurrence and astrophysical attribution

crossing, while at $5\times$ no disposition changes; transmitters drifting faster than the $12.0\text{--}13.3 \text{ Hz s}^{-1} \text{ GHz}^{-1}$ ceiling remain unconstrained. Intra-integration smearing is negligible for 434 windows: at most 1.10 channels, median 0.001. 9 windows have $\eta_{\text{smear}} < 0.99$ (Appendix A); 5 of them, all 15.3 kHz Band 6 windows and so among the most drift-capable here, carry effective thresholds $1.65\text{--}1.74\times$ nominal, and 2 are stage-1 outliers. Channel widths are effective resolutions from the measurement sets’ spectral window tables (Appendix A).

4.3. Frequency and velocity reference frames

Every frequency axis here, searched spectra, tabulated ranges and crossing frequencies alike, is *topocentric* as delivered by the ALMA correlator, with no velocity-frame shift applied at any stage. Velocities enter only in the vetting stage’s line mask, whose conversion chain is in Appendix A. The stacking is drift-aligned and inverse-variance weighted: coherent in frequency trajectory, and not in electric-field phase.

The native channel width is also the linewidth convention for every EIRP threshold, and the commonest confusion when limits are compared across surveys. Channel dilution spreads a sub-channel carrier’s total flux across the channel, so $\text{EIRP}_{5\sigma}$ is the minimum *total* power of a channel-confined carrier, independent of the transmitter’s own width below the channel, and never a spectral power density. The Hz-resolution surveys’ limits are total powers in the same sense, each the best case of its own channelisation, which is what makes the axes of Fig. 1 commensurable. We do not project an Hz-resolution sensitivity from ours: scaling by $P_{\min} \propto \sqrt{\Delta\nu}$ over six orders of magnitude would be a thought experiment, not an achievable ALMA sensitivity.

Every spectral window of an observation enters the spectral-carrier search, capped at 8 per target to bound compute cost; the cap bound one target/band, 61 Vir Band 7, and three further targets are represented by their deepest single window and marked in the released catalogue. Each searched window yields its own $\text{EIRP}_{5\sigma}$ row, and Figure 1 plots only the deepest per star. When a crossing is checked against the masked species the operative discriminant is velocity coincidence, since circumstellar emission sits at rest offset by the gas’s kinematics while a technosignature may appear at any frequency and drift rate.

Both carrier-lane thresholds and continuum fluxes are corrected for ALMA’s primary-beam response at the star’s offset from the phase centre, negligibly for most targets (median factor 1.00003, 90th percentile 1.018),

though 41 of the 443 retained windows exceed 2 per cent and 31 exceed 10 per cent. Retention is decided by an explicit response floor rather than case by case: a window is kept only where the primary-beam response at the stellar position is at least 0.5, a correction of at most $\times 2$, because at low response a small error in the beam model becomes a large error in inferred flux. *The offsets are bimodal and the floor sits in the gap*: the last retained window has response 0.55 at $0.46 \theta_{\text{PB}}$ (j1256-1257 Band 7, $1.81\times$), the first window beyond it has response 0.35 at $0.62 \theta_{\text{PB}}$ ($2.88\times$), and nothing lies between them, so the rule binds prospectively without having been drawn around the present sample. It excludes none of the 443 released windows for that reason and not because it was chosen not to. Beyond it are the 4 ϵ Eri Band 6 windows at $2.9\text{--}3.4\times$, which is also past the first sidelobe transition where the Gaussian form fails: the two criteria are independent and select the same four windows. We *withhold* them, and they enter no number here and are not in the released catalogue. Under the same nominal rule they would have covered $4.1 \times 10^{13}\text{--}2.4 \times 10^{14} \text{ W}$ at 3.22 pc, which at the lower end is among the deepest thresholds in the survey, and that is exactly why we decline to quote them on a beam model we cannot defend. The annulus geometry keeps the pipeline’s own $\theta_{\text{PB}} = 1.22 \lambda/D$, an Airy first-null coefficient; re-computing them on ALMA’s FWHM $1.13 \lambda/D$ at each window’s own dish diameter moves 33 of 443 windows by more than 1 per cent, extremes -7.3 to $+10.3$ per cent, and no headline threshold moves.

The natural benchmark here is frequency-matched and purely geometric. For a filled circular aperture of diameter D radiating P_{tx} at wavelength λ ,

$$G = \frac{4\pi A_{\text{eff}}}{\lambda^2}, \quad \text{EIRP} = P_{\text{tx}} G, \quad (4)$$

so a 12-m aperture radiating 1 MW at 230 GHz has $G = 8.4 \times 10^8$ and $\text{EIRP} = 8.4 \times 10^{14} \text{ W}$, which the median Class A effective threshold $1.8 \times 10^{15} \text{ W}$ misses by a factor 2.3. The ν^2 scaling used below is the same equation at fixed A_{eff} . Thresholds are also compared to an Arecibo-like planetary radar (Drake 1974; Sheikh et al. 2025, $\text{EIRP} 2.0 \times 10^{13} \text{ W}$), which is an 2.38 GHz power scale carrying no transmitter model and is labelled in the figures as an EIRP benchmark on the power scale only: aperture gain scales as ν^2 , so the same aperture and transmitter power at 230 GHz would radiate $2.0 \times 10^{13} \times (230/2.38)^2 \simeq 2 \times 10^{17} \text{ W}$. Because that comparison is between physical transmitter powers it must use P_{eff} and not the trigger, which carries the deepest

window from below that numerical EIRP to 3.7×10^{13} W above it (§6.1).

4.4. Ancillary and validation analyses

Nothing in this subsection generates candidates; it is validation and robustness testing, and the detail is in the appendices. The astrophysical validation is reported with the search results: the recovery and localisation of β Pictoris CO in two transitions and four blocks (§5.3.7) is the only end-to-end demonstration here that the pipeline detects a genuine narrow feature at a known stellar position. Synthetic tones of known amplitude were injected into real visibilities and recovered with the unmodified pipeline, 1200 trials across six window configurations, three bands and both classes (Appendix B); the baseline step is a per-integration median along *frequency* (Table 3, step 4) and so cannot remove a one- or two-channel feature, and the de-drift step accumulates a persistent carrier along its track. The primary statistic has no automated injection test on every pipeline change, so three quantified checks bound undetected faults instead: the machinery-only lane, amplitude fidelity $-9/+14$ per cent; the AU Mic re-analyses, whose recomputation agrees with the released products; and pseudo-star rank uniformity over 220 672 pooled ranks ($p = 0.37$).

4.5. Dwell-fraction completeness

How long a carrier occupies one native channel defines the searched class, so we measure recovery against that directly. A carrier was injected into the retained per-integration spectra of 12 real windows, six coarse and six fine, at amplitudes of 2–20 times the per-integration noise and dwell fractions $f_{\text{dwell}} = 0.10$ –1.00, six realisations each: 3888 trials, released with one row per injection (Table 4). Its detection criterion, recovered when the logged per-integration S/N reaches five, was re-derived from the released rows and agrees with the logged flag 3888 times in 3888 trials, with 0 mismatches across a clean separating gap. Recovery rises monotonically with dwell. Pooling amplitudes at or above 4σ , coarse windows recover 100 per cent of continuous carriers, 100 per cent at $f_{\text{dwell}} = 0.25$ and 83 per cent at 0.10; fine windows give 100, 91 and 71 per cent.

An end-to-end check confirms this through the unmodified pipeline. Injecting a zero-drift, always-on tone into the calibrated visibilities of a coarse Band 7 window at 1, 3 and 10 times the per-integration rms returns $T_\star = 20.5$, 65.3 and 217.7, each within one channel of the injection. The ideal stacking gain over its 639 integrations is $\sqrt{639} = 25$, and the measured 20.5 is that value less weighting and flagging losses.

Two distinctions apply. The dwell campaign injects into retained per-integration spectra, after the baseline step, while the end-to-end check injects into visibilities before it; and one cell, the through-pipeline recovery of near-static carriers on Class A windows, is measured only in part.

The campaign measures completeness at the survey threshold and across configurations. The trigger acts on the stacked statistic and the de-drift step is an inverse-variance-weighted sum over integrations, so a carrier of per-integration amplitude A present for a dwell fraction

TABLE 4

COMPLETENESS MATRIX. (a) WHAT IS MEASURED FOR EACH SEARCHED MORPHOLOGY AND WHERE: ‘e2e’ IS INJECTION INTO VISIBILITIES THROUGH THE UNMODIFIED PIPELINE, ‘POST’ INJECTION INTO RETAINED PER-INTEGRATION SPECTRA AFTER THE BASELINE STEP, AND ‘CFG’ THE NUMBER OF WINDOW CONFIGURATIONS THE CELL SPANS. THE TWO CAMPAIGNS MEASURE DIFFERENT QUANTITIES: 1200 INJECTION TRIALS (APPENDIX B) AND 3888 DWELL TRIALS OVER 12 REAL WINDOWS.

Morphology	e2e	cfg	thresh.	dwell	drift
Class A drifting	yes	1	yes ^a	no	yes
Class A near-static	post	6	no ^b	yes ^c	n/a ^c
Class B stationary	yes	6	yes ^d	yes ^e	none
Class B drifting	yes	6	yes ^d	yes ^e	none ^c
Intermittent, either class	post	12	no	yes ^e	by class

^a42 per cent at the trigger, 83 at twice it, one configuration only (AU Mic, Band 6, 488 kHz), transfer to the other 126 Class A windows measured at 0.64–1.13 $\times$ on 28 configurations (Fig. 7).

^bAfter the baseline step only. ^cSub-channel over the track by definition. ^dHalf recovery at stacked $a = 5.3$ –8.3 in the 6 windows straddling the trigger. ^e83–100 per cent (coarse), 71–100 per cent (fine), over $f_{\text{dwell}} = 0.10$ –1.00.

(b) completeness at the trigger, against stacked amplitude a (Eq. 5), pooled over the 12 injected windows; trial counts in parentheses.

stacked a	Class B (coarse)	Class A (fine)
< 3	0% (30)	0% (72)
3–4	0% (12)	7% (42)
4–5	3% (30)	12% (42)
5–6	46% (24)	33% (48)
6–8	58% (60)	70% (96)
8–12	100% (108)	99% (174)
> 12	100% (1680)	100% (1470)

f_{dwell} of a track of N_{int} integrations accumulates to

$$a = A f_{\text{dwell}} \sqrt{N_{\text{int}}} \quad (5)$$

in units of the per-integration noise. Recovery depends only on the combination a , so binning the released trials in a unifies the two axes and places all 12 injected windows on one axis. Those windows span a factor of 63 in integration count over 11 targets, two resolution classes and Bands 3, 6 and 7, so configuration-independence is measurable. In the 6 windows whose amplitude grid straddles the trigger, the 50 per cent recovery point lies at $a = 5.3$ –8.3. That is the 5σ threshold itself, independent of channelisation, integration count and band. In the other 6 the weakest injection already stacks to $a = 3.4$, so they are consistent but uninformative about the curve’s position. The pooled curves (Table 4) agree between classes within the sampling noise.

The injected carriers do not drift, so what is measured across 12 configurations is the persistent and partial-dwell non-drifting class. On Class B windows that is the searched class, which therefore carries a survey-threshold completeness curve; on Class A windows the searched class is the drifting carrier, whose curve is still the single calibration configuration of Appendix B, so the fine column of Table 4 bounds nothing about the transfer error. The quantity a is a derived combination, justified by the form of the statistic and by the collapse of 12 configurations onto one curve rather than measured independently at every (A, f_{dwell}) pair.

One caveat attaches to the drift ceiling. The searched value is 12.0–13.3 Hz s^{−1} GHz^{−1}, line-of-sight $|a| = 3.60$ –4.00 ms^{−2}, the generic value everywhere except the 12 windows where a known planet’s own bound raises it.

It clears Earth-rotation and Earth-orbit drifts by two orders of magnitude but is marginal against a close-in planet on a TRAPPIST-1 b-like orbit, so the limits of §6 are conditional on it (Figure 8).

Interference rejection, which is an operative gate. Satellite constellations grew across the 2013–2025 archival baseline, and although no ordinary downlink allocation maps into the searched frequencies, an allocation table excludes neither harmonics nor intermodulation products nor unconventional emitters, so no exclusion claim here rests on one. The defences that do the work are the control ensemble and the time domain. A satellite enters the star and the control extractions alike within an integration; the mismatch between its bandwidth and ours, together with the drift-stacked sum over the track, dilutes a transient pass; and a second epoch at the same tuning tests whether anything reproduces. What none of that reaches is a near-field emitter, whose interferometric response need not resemble a celestial source’s at all, which is one reason a per-integration RFI screen and a visibility-domain check belong in any successor analysis.

What follows is the final analysis. The pipeline as released differs from earlier working versions in five substantive ways, logged with their effect on released products in Appendix G; only the one §5.4 dates changes a released number, and asymmetric-derived quantities are historical there.

Scope of the constraints. The spatial screen is empirically calibrated, not exactly exchangeable, and the radial correction of §5.3.2 is derived on the calibration sample and therefore *not validated out of sample*; no disposition here depends on it. *What was searched*, by class. Class A, the fine-channel search: 25 frequency islands totalling 43.2 GHz between 114.0 and 873.1 GHz, channels 0.02–1.95 MHz, line-of-sight accelerations $|a| \leq 3.6\text{--}4.0\text{ m s}^{-2}$. Class B, persistent channel-confined excess with no drift discrimination: 36 islands totalling 89.9 GHz between 89.6 and 682.8 GHz, channels 15.62–31.25 MHz. Both: only the archival epochs observed, only carriers one native channel or narrower, and 109.4 GHz of the 113.9 GHz union after molecular-line masking, at nominal trigger thresholds whose measured recovery is 42 per cent, 27–47 per cent under the measured $0.64\text{--}1.13\times$ transfer. Inside that box the non-detection is a constraint. *Nothing in this paper bounds the prevalence of:* (i) drift-rate structure on the 317 coarse windows, which carry no drift discrimination and no survey-threshold completeness curve (§4.5); (ii) transmitters emitting *outside the searched spectral windows*, meaning those 45 islands, not ALMA’s frequency range (Fig. 3); (iii) transmitters active *outside the archival epochs*: the limits assume near-continuous emission and are only weakly constraining for intermittent transmitters; (iv) drift rates *above* the searched ceiling $|a| = 3.60\text{--}4.00\text{ m s}^{-2}$, excluding the most close-in orbits (Fig. 8); (v) carriers placed *inside* the $\pm 50\text{ km s}^{-1}$ molecular-line mask, which is excluded search space and not searched-and-empty; (vi) emission *broad*er than one native channel, for which the thresholds must be rescaled; (vii) anything below the nominal threshold, where measured recovery is 42^{+16}_{-14} per cent (Appendix B); (viii) any transmitter-frequency prior spanning ALMA’s tuning range, under which the conditional bound vanishes (§6.2); (ix) *harmonics and inter-*

TABLE 5
THE SURVEY AS SEARCHED, AND THE PAPER’S HEADLINE NUMBERS IN ONE PLACE. EVERY COUNT QUOTED IN THIS PAPER RECONCILES WITH IT. THE TWO COMPLETENESS EXPERIMENTS MEASURE DIFFERENT QUANTITIES AND ARE KEPT IN SEPARATE TABLES.

Quantity	Value
Stars within 40 pc with qualifying public ALMA coverage	168
genuinely pointed at; never observed (§5.2)	115; 53
Stars searched; independent systems	88; 81
Star/band datasets	109
Spectral windows searched	443
12 m array; ACA 7 m extracted; repeats, defective, withheld	302; 141 474; 21, 6, 4
Class A fine-channel / Class B coarse-channel	126 / 317 (72% Class B)
Execution blocks	104
Distance range	1.30–39.63 pc
Unique frequency union	113.9 GHz (45 intervals, 89.6–873.1 GHz)
Windows by ALMA band (3/4/5/6/7/8/9/10)	40/4/4/216/155/12/4/8
Effective after line-exclusion mask	$\approx 109.4\text{ GHz}$
Native channel widths	15.3 kHz–31.25 MHz; median 15.625 MHz
Drift-rate ceilings	1.1–5.9 kHz s ^{−1} (12.0–13.3 Hz s ^{−1} GHz ^{−1})
On-source time per window	21–5274 s (median 2087 s)
Nominal $S_{\min}$	0.48 mJy–193.17 Jy (median 6.9 mJy)
Nominal trigger P_{trig} , Class A (15.3–1953 kHz)	$2.3 \times 10^{13}\text{--}6.8 \times 10^{16}\text{ W}$ (median 7.9×10^{14})
Nominal trigger P_{trig} , Class B (15.625–31.25 MHz)	$1.6 \times 10^{13}\text{--}2.8 \times 10^{17}\text{ W}$ (median 2.2×10^{15})
Effective $P_{\text{eff}} = 2.29 P_{\text{trig}}$, Class A	$5.4 \times 10^{13}\text{--}1.6 \times 10^{17}\text{ W}$ (median 1.8×10^{15})
Class B	$3.7 \times 10^{13}\text{--}6.4 \times 10^{17}\text{ W}$ (median 5.1×10^{15})
Completeness, by class and experiment	Table 4 (amplitude); Table 4 (dwell)
Threshold crossings ($\geq 5\sigma$ at the stellar position)	20 windows (all Class A)
Star-exceeds-ring windows ($\geq 5\sigma$)	4 windows (all Class A)
circumstellar or foreground CO	3
unattributed, absent in the repeat block	1
Credible technosignatures	0

modulation products, which a search keyed to one channel cannot assemble across a spectrum; (x) *polarisation*: the search is Stokes- I only. All 104 of the 104 execution blocks deliver both parallel hands and 0 deliver a cross-hand product, so no Stokes Q , U or V is recoverable from the archived data for any searched window. Total power is unaffected, since a fully polarised carrier appears in I at its full strength, but a per-hand search would gain up to $\times 1.41$ in signal-to-noise on such a carrier, and if one hand were lost to flagging in a window we could not tell, which is worth a factor two in power in the worst case.

5. RESULTS

One late correction stands behind the non-detection and it made the picture more conservative: the mid-analysis statistic revision only ever removed outliers, seven windows before and four after (§5.4, Table 9).

Per-target values, including each band’s searched frequency range, are in the released catalogue (Data Availability).

5.1. Barnard’s Star and Wolf 359

Barnard’s Star and Wolf 359 are the second- and fourth-nearest stellar systems. We find no prior mm or

submm technosignature limit for either, so these appear to be the first. Both are non-detections in public Band 6 archival data. The nominal EIRP_{5 σ} trigger thresholds are $6.2\text{--}6.9 \times 10^{13}$ W for Barnard’s Star over four windows and $7.8\text{--}9.1 \times 10^{13}$ W for Wolf 359, with continuum upper limits of 0.575 and 0.453 mJy.

5.2. Data-quality exclusions and open items

109 target/bands are searched to completion and reported here. Four classes of entry are absent from the released catalogue and are itemised in Appendix G.

A further 68 candidate target/bands were selected but never delivered a searchable product, and the cause lies in the selection step. Re-running the crossmatch shows that 60 of them were never observed at all, no observing unit offered to those entries containing the star: they sit a median 1.6 degrees from the nearest field centre, because the observations were pointed at solar-system targets whose quoted field of view is enormous, and a crossmatch that trusts that quantity admits stars far outside any real primary beam. The remaining 8 entries did not yield a search product. A positional crossmatch must not use the quoted field of view of an ephemeris or solar delivery as a search radius; the per-star pointing ledger is released so the selection can be audited. 53 of the 168 stars admitted to the candidate list fail on pointing in every band, so the sample here is 88 of at most about 115 genuinely covered stars rather than 88 of 168, and the coverage completeness of Table 5 should be read against the smaller denominator.

5.2.1. The coarse-noise defect, and the bound on its reach

A reproducible defect produces implausibly small noise in a minority of coarse-channel windows. The radiometer relation $\sigma \propto \text{SEFD}/\sqrt{N_{\text{bl}}\Delta\nu_{\text{ch}}t_{\text{on}}}$, in the system equivalent flux density (SEFD), makes $q \equiv \sigma\sqrt{t_{\text{on}}\Delta\nu_{\text{ch}}} \propto \text{SEFD}/\sqrt{N_{\text{bl}}}$ a quantity physics predicts, so a window whose q lies far below the sample median is too extreme for a noise fluctuation and points instead to a bookkeeping defect; we exclude such windows at $100\times$ below the median. Six of 443 windows fail, across five stars. Their nominal thresholds of $2.0\text{--}7.8 \times 10^{12}$ W would otherwise have set this paper’s headline depth. Excluding them, the deepest is 1.6×10^{13} W.

The exclusion does not rest on the retained population being narrow, since q runs over a factor ~ 290 of the sample median, mostly the band dependence the same relation predicts through SEFD. It rests on the size of the gap and on the direction of the error. The six sit a factor 1.5×10^3 below the lowest retained window across an empty 3.2 decades, so any threshold between 1.5×10^1 and 2.2×10^4 in absolute q selects the same six, and modelling q for every retained window from its own band, array and antenna count leaves the residuals unimodal with the six 3.0dex below the lowest retained window and nothing between. Direction settles the rest: σ is estimated from the control probes and applied to every position of a window alike, so a window-level deflation can only over-state sensitivity and can never manufacture a star-exceeds-ring window.

The six share a signature. They are all 12m array, 3 in Band 6 and 3 in Band 7, all at 15.6 MHz, in 5 execution blocks across 4 projects, and every one has a normally

behaved finer-channelised window covering the same frequencies in the same block: a channel-averaged auxiliary window riding on a frequency-division science baseband, carrying a constant WEIGHT column and so never given that baseband’s calibration (Appendix G). The deflation is common mode in the data and in the weights alike, so it leaves the signal-to-noise ratio the search uses untouched, which is why the defect can misstate a threshold, cannot manufacture a flag, and costs no frequency coverage at all. The affected windows are therefore excluded from the sensitivity accounting only: their signal-to-noise statistics and ranks remain valid, and were carried through candidate generation on that basis.

21 of the 474 extracted rows are duplicate catalogue entries for one physical window, the same (star, execution block, frequency range, channel width) key appearing twice in the archive query, so one row per key is the right statistical unit. The retention rule is deterministic and quantity-independent, the key’s first-listed row in the query’s own identifier order, so it consults no measured value and cannot prefer a crossing or a deeper threshold. Repeats need not agree: two reductions of the same τ Ceti Band 6 window differ by 44 per cent in threshold, which exceeds the flux-scale systematic of §4.3 and is the most direct empirical bound here on threshold reproducibility, and the two HR 1010 Band 6 rows straddle 5σ at 5.10 and 4.95, so the crossing count is rule-dependent at the level of one window. The four flagged windows are not, no discarded row being both a crossing and a stage-1 spatial outlier (Appendix G).

5.3. Technosignature search

All 109 star/band datasets yielded at least one carrier-lane result, 443 spectral-window measurements after the catalogue audit removed 21 duplicate rows, Bands 6 and 7 dominating (Table 11). Nominal trigger thresholds span $1.6 \times 10^{13}\text{--}2.8 \times 10^{17}$ W at a median of 1.5×10^{15} W (Figs. 3 and 1), the deepest being reached toward the nearest systems, 1.6×10^{13} W for the UV Ceti pair at 2.7 pc, though depth also tracks integration time and channelisation.

The statistics come strictly before the astrophysics, in three named stages that do not promote one another. A *crossing* is one channel $\times$ drift cell at the stellar position reaching $T \geq 5$: 20 of the 443 windows contain one, and all 20 are Class A. That is a trials effect rather than a physical one: a median Class A window holds 2.5×10^5 channel $\times$ drift cells against 351 for a median Class B window, a factor 714, so a 5σ cell is correspondingly rarer in the coarse class. A *stage-1 spatial outlier* is a window whose stellar statistic also exceeds all 512 controls: four do. That screening device selects a window for examination (§4.2). Each of the four is then carried past that floor by a local scramble test ((v) below), which asks whether a feature exceeds the noise at its own position without discriminating spatially; in 3 of the 4 windows the control-ring maximum clears it too, so candidacy still turns on the first-stage star-versus-control comparison. A *candidate* is a stage-1 outlier surviving every veto, and there are none. Three of the four have independent CO evidence, matched transitions, velocities and epochs at β Pictoris and published foreground emission at HD 48370; the fourth, towards CP–72 2713, never

acquired one and is an unconfirmed first-block event, absent from a repeat block, below. Those three are disposed of by astrophysical evidence, with the $\pm 50 \text{ km s}^{-1}$ mask playing no part; CP-72 2713 has no astrophysical attribution and fails the persistence criterion instead, which excludes a persistent emitter of that strength but not an intermittent one. Table 6 gives each flagged window with the evidence dispositioning it.

5.3.1. The external calibration sample

The statistic and the mask were settled on the 443 windows above, so they cannot also calibrate themselves. A separate set of 150 execution blocks toward 27 stars, 603 windows, was run through the same frozen pipeline with no change to statistic, mask, threshold or candidate rules. 23 of those blocks are toward 11 stars that carry no flag in the survey, so this is not a re-test of the flagged windows. We call it a calibration rather than a validation sample: it measures the statistic’s behaviour out of sample, and the radial correction of §5.3.2 is derived on it, so it cannot also validate that correction.

Table 7 is the accounting. 4 windows carry β Pictoris CO at the stellar position, which is what a working statistic should do where a real localised signal is present. 20 more cross the trigger at the star, but the control annulus crosses it too, reaching $T = 39$: that is resolved emission filling the field, not a spatial outlier. One window, toward HD 23484 (Gaia DR3 4856592239127350400) in Band 6, is a stage-1 outlier with no astrophysical attribution, and the same tuning in 3 further blocks of that star shows no repeat.

Part (b) of Table 7 sets it beside CP-72 2713. The two events are the same kind of thing: a single cell a few per cent above the largest of 512 controls, in a window whose repeat observations put the stellar position back in the body of its own control distribution. One arose in the sample that defined the statistic and one in a sample that could not have, which is the plainest demonstration we can offer that stage-1 status carries no candidate significance on its own, and that recurrence, not rank, is what settles a marginal event.

The rank distribution of the remaining 578 windows is the campaign’s most consequential result, and it is adverse to the method. The star’s add-one rank is not uniform. Its median is 0.44 where 0.5 is expected, 14 per cent of the ranks fall below 0.1 against 10 per cent expected and 6 per cent above 0.9 against the same, so the whole distribution is displaced towards the star rather than growing a tail. Windows inside a block share a calibration and blocks of a unit set a target, so the window-level figure ($D = 0.08$, $p = 0.00$) treats as independent what is not; clustering removes neither the shift nor its significance, 99 of the 149 blocks having a mean rank below 0.5 (sign test $p < 0.001$) and a star-clustered bootstrap placing the median at 0.44 (95 per cent interval 0.42–0.46). It is not peculiar to the campaign either. The survey’s own 443 windows lean the same way without resolving it, median 0.47 against a star-clustered 95 per cent interval of 0.42–0.51, and the two distributions are statistically indistinguishable ($D = 0.08$, $p = 0.05$), so the in-sample checks above were underpowered against a shift of this size rather than contradicted by it. **The control ensemble is therefore an empirically calibrated screen for extreme values and not an ex-**

actly exchangeable rank test, and the rest of this subsection is about which of those two statements the null result needs.

5.3.2. Where the displacement comes from: radius, not stars

The control probes are not redrawn per window. The pipeline draws them from one fixed seed as $r = \sqrt{U(r_{\text{in}}^2, r_{\text{out}}^2)}$ with $r_{\text{in}} = 0.14 \theta_{\text{PB}}$ and $r_{\text{out}} = 0.78 \theta_{\text{PB}}$, so the fractional radii are identical in every window of the survey and each stored control statistic can be paired with the radius at which it was measured. The star is not at a comparable radius: its median offset from the pointing centre is $0.0035 \theta_{\text{PB}}$, an order of magnitude inside the annulus’s inner edge. Three tests follow, all from the released products and none needing a new search.

First, the control level itself depends on radius: in units of each window’s own robust noise scale the pooled control statistic runs $+0.20$ at $u = 0.24$, through a minimum of $+0.10$ near mid-annulus, back to $+0.14$ at the outer edge (calibration: $+0.15$, $+0.07$, $+0.18$), so positions near the pointing centre sit high before any star is involved. Second, radius matching removes most of the departure: ranking the star against only the 54 innermost probes ($u \leq 0.30$) moves the survey median from 0.455 to 0.473 and $D = 0.066$ ($p = 0.06$) to 0.050 ($p = 0.28$); the 35 windows whose star happens to lie inside the annulus have median rank 0.489 against 0.450 for the 345 that do not. Third, treating an innermost probe as a *pseudo-star* and ranking it against the other 512–1 puts a position containing no star through the same test: out of sample the pseudo-stars are displaced as far as the real stars, median 0.443 against 0.429 ($p = 0.003$), though in the survey they are not (0.506 , $p = 0.71$).

The displacement is therefore a property of *position in the field* and not of stars, which is what the same displacement would have meant had the pseudo-stars come out uniform.

5.3.3. The survey rescored with a repaired screen

The candidate generation was rerun under two corrections, both built from the stored control vectors and starting no new search. The first is *radius matching*: rank the star against only the 54 probes with $u \leq 0.30$, the closest approach to an iso-primary-beam comparison the frozen products allow. Its cost is resolution, the ensemble falling from 512 probes to 54, so the screen admits 8.1 windows by chance alone; it returns 5, of which 1 is new. The second keeps all 512 probes and removes the trend instead: with $m(u)$ the pooled radial profile measured *on the calibration windows*, subtract $sm(u_k)$ from each control and $sm(0)$ from the star, s being that window’s robust scale. Rank resolution is unchanged at $1/(512 + 1)$. This is the repair we adopt, because the trend is a property of the field rather than of the star: pseudo-stars, control positions carrying no star at all, are displaced as far as real ones (§5.3.2). Removing a measured profile therefore uses every probe, where radius matching discards nine in ten of them to approximate the same correction.

The debit $m(0)$ is the one free choice. Taking the innermost measured bin, 0.15σ , is conservative, because the profile is still rising inwards there. Taking instead the value that centres the calibration-sample rank distribution on 0.5 gives 0.32σ . Rather than argue for

TABLE 6

THE FOUR STAGE-1 SPATIAL OUTLIERS, WITH THE EVIDENCE DISPOSITIONING EACH. T_* IS THE SEARCH STATISTIC AT THE STELLAR POSITION (EQ. 2) AND ‘RING MAX’ THE LARGEST OF THAT WINDOW’S 512 CONTROL STATISTICS; ν_{cross} IS THE CROSSING CHANNEL, WHICH DIFFERS FROM THE WINDOW CENTRE BY UP TO 0.85 GHz HERE, AND $S_{\text{peak}} = T_*\sigma$ IS THE PEAK FLUX DENSITY IN ONE NATIVE CHANNEL. $\Delta\nu$ AND Δv GIVE THE TOPOCENTRIC OFFSET OF THAT CHANNEL FROM THE NEAREST *laboratory* REST FREQUENCY OF THE MASKED SPECIES, SIGNED SO THAT A NEGATIVE VALUE PLACES THE OBSERVED CHANNEL *below* THE REST FREQUENCY, THAT IS, REDSHIFTED RELATIVE TO IT. THE MASK OF §5.3.10 CHANGES NO ROW, AND THE HD 48370 ROW IS THE LIMITING CASE, A NEAR-TIE OF RANK DISPOSITIONED BY VELOCITY COINCIDENCE WITH INDEPENDENTLY PUBLISHED EMISSION.

Window	ν_{cross} (GHz)	T_*	ring max	S_{peak} (mJy)	$\Delta\nu; \Delta v$ (MHz; km s ⁻¹)	Disposition
β Pic B3	115.260644	14.65	7.27	54	-10.56; -27.5 (CO 1→0)	circumstellar CO
β Pic B6	230.516128	11.68	9.12	78	-21.87; -28.4 (CO 2→1)	circumstellar CO
HD 48370 B6	230.511674	27.10	26.83	0.82×10^3	-26.33; -34.2 (CO 2→1)	foreground CO
CP-72 2713 B7	344.269746	5.81	5.68	12	-1526.24; -1323.2 (CO 3→2)	unattributed; no repeat

TABLE 7

THE OUT-OF-SAMPLE CAMPAIGN. (a) WHAT THE FROZEN PIPELINE RETURNED ON 603 WINDOWS IT HAD NEVER SEEN, BY CATEGORY. (b) THE SURVEY’S ONE UNATTRIBUTED STAGE-1 OUTLIER AND THE CAMPAIGN’S, SIDE BY SIDE: BOTH ARE SINGLE CELLS A FEW PER CENT ABOVE THE LARGEST OF 512 CONTROLS, AND BOTH FAIL TO RECUR AT THE SAME TUNING. RANKS ARE ADD-ONE RANKS OF THE STAR AMONG ITS CONTROLS.

(a) Campaign, as searched

Execution blocks / stars	150 / 27
Spectral windows	603
β Pic CO at the star ($T_* = 8.9$ –30.8)	4
crossing at the star, annulus above it too	20
stage-1 outlier, no attribution	1
effectively null	578
Stage-1 outliers among the null windows	0

(b) The two marginal outliers

	CP-72 2713	HD 23484
Sample	frozen survey	calibration
Band	7	6
ν (GHz)	344.269746	230.790
T_*	5.81	5.30
largest control	5.68	5.25
99th control percentile	5.57	5.10
Repeat blocks at the same tuning	1	3
T_* in the repeats	3.60	4.14–4.99
largest control in the repeats	5.93	5.49–5.67
rank in the repeats (of 513)	188	21, 219, 381

one, Table 8 reports how large a debit each flagged window survives. The two β Pictoris windows survive 10.80 and 25.77 σ , twenty to eighty times either choice. CP-72 2713 survives 0.61 σ , twice the calibrated debit. HD 48370 survives only 0.21 σ : its stellar peak stands 0.27 above a ring maximum of 26.83, because the emission is extended and fills the annulus, which is exactly why it is dispositioned as foreground CO.

No window is promoted by either repair. Under the adopted screen the flagged set is the released one, 4 windows, with 0 added and 0 lost. Under radius matching the one addition is a Class A window at $T_* = 5.11$, against 8.1 expected by chance at that screen’s resolution. Under the strongest debit the set loses HD 48370, an attributed line, and keeps the rest. The number of *unattributed* stage-1 outliers is one for every repair and every debit between zero and 0.61 σ , so no disposition in this paper depends on the defect.

What the repair does not do is make the rank distribution uniform. Detrending moves the calibration-sample median from 0.429 to 0.437 ($D = 0.091$ to 0.081) and the survey median from 0.466 to 0.474 ($p = 0.32$ to 0.60): radius is part of the story and not all of it. Nor would fitting and testing the profile on the same windows validate it. Leaving each of the 27 calibration stars out of the fit in

TABLE 8

THE SURVEY RESCORED UNDER BOTH REPAIRS. “SURVIVES” IS THE LARGEST STELLAR DEBIT $m(0)$, IN UNITS OF EACH WINDOW’S OWN CONTROL SCALE, AT WHICH THE STAR STILL EXCEEDS EVERY DETRENDED CONTROL; IT IS A PROPERTY OF THE WINDOW AND ASSUMES NEITHER DEBIT. FOR REFERENCE THE CONSERVATIVE DEBIT IS 0.15 σ AND THE DEBIT THAT CENTRES THE CALIBRATION-SAMPLE RANK DISTRIBUTION IS 0.32 σ , AND A WINDOW SURVIVES WHICHEVER DEBIT IS SMALLER THAN ITS ENTRY. NO WINDOW IS ADDED BY EITHER REPAIR.

Window	T_*	Survives (σ)	Rad.-m.	Detr.
β Pic B3	14.65	25.77	yes	yes
β Pic B6	11.68	10.80	yes	yes
HD 48370 B6	27.10	0.21	yes	yes
CP-72 2713 B7	5.81	0.61	yes	yes
Flagged, whole survey			5	4
Added			1	0
Expected by chance			8.1	0.9

turn and scoring it against a profile it did not contribute to returns median 0.437 and $D = 0.081$, against 0.437 and 0.081 in sample: the profile is not over-fitted, and it does not restore exchangeability either. We therefore do not present the detrended statistic as a validated null. It is a robustness check on the dispositions, and the survey’s false-alarm budget is taken from the measured tail rate below instead. A parametric beam model would not settle it: the windows mix 12-m and 7-m primary beams (Appendix F), and a displacement that survives where no star is present is not the beam acting on a star. The residual is why §6.4 puts visibility-domain localisation, which compares no sky positions at all, ahead of any further repair of a position-ranking statistic.

5.3.4. The false-alarm rate, measured instead of assumed

Because the body of the rank distribution is displaced, $1/(512+1)$ is the rank *resolution* of the ensemble and not a per-window null probability. The survey’s false-alarm budget is therefore quoted from measurement. Two independent measurements agree. Of the 599 calibration windows carrying no attributed line, 1 exceeds all 512 of its controls, one in 599 or 0.9 times exchangeable; and the pseudo-stars, which cannot contain a signal at all, rank first in 0.28 per cent of trials out of sample against 0.19 per cent exchangeable, a factor 1.4. The uncertainty on that factor is not binomial: the 16 inner probes of one window share its noise realisation and its execution block. Recomputing the test from the stored control vectors over all 603 calibration windows gives 30 first ranks in 9648 trials, a factor 1.6, and resampling whole windows, or whole execution blocks (150 of them), widens the 95 per cent interval barely at all: 1.1–2.2 and 1.1–2.1 against 1.1–2.3 binomial. The reconstruction and the re-

leased value differ (1.6 against 1.4) only in whether the four line-carrying calibration windows are included, and both sit well inside the interval. The factor is therefore established as lying between about 1 and 2 rather than to two figures, which is the precision the argument needs: no disposition in this paper changes anywhere in that range. The tail is *mildly* anti-conservative where the body is *strongly* displaced, and it is the tail the candidate screen uses. Propagated over the survey's 443 windows, the pseudo-star rate predicts 1.2 unattributed stage-1 outliers, with $P(\geq 1) = 71$ per cent, against 0.86 predicted under exchangeability and one observed. Every per-window probability quoted in this paper should be read as multiplied by 1.4, which changes no disposition here: no window was claimed significant after the survey-wide correction of Appendix F, and CP-72 2713 is expected at this rate rather than surprising under it. The displacement is widespread rather than one configuration's doing, 23 of 27 stars having a median rank below 0.5 (Fig. 4).

The campaign covers 27 stars in three bands and windows within a block are not independent epochs, so the sizes above carry that caveat; at 578 null windows the test resolves a shift of this size ($D \geq 0.06$ against 0.08 observed). On data that played no part in building it the frozen pipeline recovers the signal present, raises no candidate, and ranks the stellar position above its annulus more often than exchangeability predicts. Counts are those at this build date (2026 September 13, 07:44 UTC).

5.3.5. Does the stellar position carry its own noise?

The noise scale of Eq. 2 is built across the annulus, so by construction it carries no variance peculiar to the stellar position, where residual continuum, deconvolution or self-calibration residuals and position-correlated calibration error would concentrate. That is the largest untested assumption in the method, and the retained products bound it. R_σ , the star's robust residual scale over the four stage-1 windows against the median of the 8 control positions the release keeps (Appendix G), runs 0.995 to 1.001, so the star agrees with its ring to better than 0.7 per cent at 95 per cent confidence. That matters most for CP-72 2713. Because $T_\star$ scales inversely with the noise the star is divided by, an unmodelled broadband excess of 2.2 per cent at the stellar position would bring that window's $T_\star$ below its own ring maximum and dissolve the outlier; the bound is a factor of three smaller, so that explanation is disfavoured without being formally excluded. The same test on a seeded random sample of 25 *unflagged* windows returns 0.997 (range 0.968–1.004), every one within 3 per cent of unity, so the no-excess-at-the-star assumption holds generally and not only where a disposition needed it. It is a test of variances and not of means, and it is blind to the one- or two-channel artefact the search is built to find, which the local null and recurrence address instead.

5.3.6. Survey-level false-alarm calibration

The per-window rank probability, its survey-level counterpart, the pseudo-star and stratum tests that show the control ensemble behaves as an ensemble, and the trials budget in cells are all calibration rather than result, and referee comment moved them together into Appendix F. What the main line of argument needs from them is the

two statements above: the tail rate is measured, and no window in this survey approaches the survey-wide correction.

5.3.7. The positive control: β Pictoris CO

A search that finds nothing must show that it could have found something, and this is where that is done. β Pictoris is the survey's *positive control*: a real, spatially localised, narrow spectral feature at a stellar position, recovered blind by the frozen pipeline, in two transitions, in four execution blocks, across nine years, and at a drift consistent with none: the Band 3 crossings agree across blocks to 1.00 channel once each block's own tuning is removed, which bounds any apparent drift far below the grid the window searches. HD 48370 is the complementary case, an extended foreground line that fills the annulus and so is rejected by the same spatial statistic that promotes β Pictoris; CP-72 2713 and HD 23484 (Table 7) are the third regime, marginal single cells that do not recur. The three together are what the method looks like when it works.

Two of the four flagged windows are towards β Pictoris, a debris disc with resolved CO emission (Matrà et al. 2017; Dent et al. 2014): one in Band 3 and one in Band 6 (Table 6). The Band 6 window is a line tuning, 53.9 MHz wide at 15.26 kHz, so it holds 3533 channels like a wide window and searches 1944 trial drift rates against a survey median of 4, the largest trials load in the survey at $\eta_{\text{drift}} = 422$: this rank is harder to earn than a typical one, not easier. Both windows sit at the same place in velocity. Carried through the frame chain, the Band 3 crossing lies $+0.44$ and the Band 6 crossing -0.29 km s^{-1} from the stellar systemic velocity, in the two lowest CO rotational transitions of a star with mapped circumstellar CO, and a third, unflagged window six years earlier shows the same component at -0.46 km s^{-1} . All three lie within 0.46 km s^{-1} of systemic. We attribute both crossings to this kinematically offset CO. The full frame audit, the flux ratio and the excitation argument are in Appendix G, with the per-window numbers in Table 15.

Attributing the pair to CO does not diminish it. Against the survey background the Band 3 peak $T_\star = 14.65$ is matched or exceeded by 5 of the 443 per-window control maxima and the Band 6 peak $T_\star = 11.68$ by the same number, add-one $p = 0.014$ and 0.014 . The alternative reading needs a transmitter placed by coincidence at matched offsets from two CO transitions in one star. The outliers also concentrate where the astrophysical foreground is richest: β Pic, AU Mic and CP-72 2713 are all β Pictoris moving-group members with debris discs, and HD 48370 is a disc host behind a molecular cloud (§6.3).

5.3.8. Dispositions of the remaining outlier windows

The third stage-1 outlier, towards HD 48370 in Band 6, is the largest excess here in absolute terms and the smallest in relative ones, $T_\star = 27.10$ against a control-ring maximum of 26.83. Star and ring rising together is what emission filling the primary beam looks like. The velocities agree immediately: Cataldi et al. (2023) report prominent CO(2→1) towards HD 48370 at $v_{\text{bary}} \approx 41 \text{ km s}^{-1}$ and flag it as possible foreground cloud contamination, and our own frame chain returns $v_{\text{bary}} =$

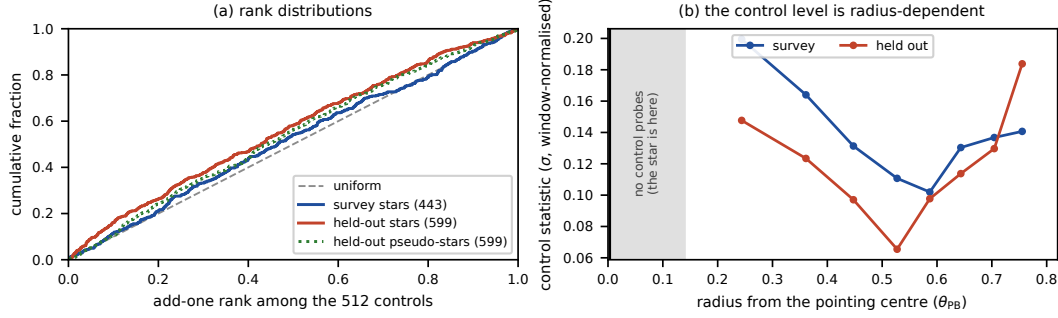


FIG. 4.— Why the spatial control ensemble is a calibrated screen and not an exchangeable rank test. (a) add-one rank of the star among its 512 controls, for the survey and for the calibration windows carrying no attributed line: both lie above the uniform diagonal. The dotted curve is the *pseudo-stars*, control positions at the inner edge of the annulus ranked against the other controls, which contain no star and are displaced just as far (§5.3.2), so the displacement is instrumental. (b) the control statistic itself against radius from the pointing centre, in units of each window’s own noise scale. It is lowest at mid-annulus and rises towards both edges; the shaded strip inside $0.14 \theta_{PB}$ holds no control probe at all, and that is where the star sits (median $0.0035 \theta_{PB}$).

+41.2 km s^{−1} for this crossing, so we recover their line and confirm it kinematically. Three checks close the circumstellar alternative. The same authors report the disc itself undetected in CO and in carbon. The sight line runs through the outer Galaxy, where a flat rotation curve reaches the measured velocity only at a kinematic distance of 2.07 kpc, which nothing local could produce. And the block’s ¹³CO(2→1) window has a control maximum of 20.4 against 4.6 at the star, so the field carries bright ¹³CO that the star does not, resolved 1.7 synthesised beams away. Appendix G gives the Band 8 [C I] tuning check, which says the same thing by omission. The rank statistic identifies nothing here, a stated limitation of the spatial test.

5.3.9. CP–72 2713: an unconfirmed first-block event, absent in the repeat block

This subsection is a worked example of how a marginal event is dispositioned, not a special object: at the measured false-alarm rate of §5.3.4 the survey expects 1.2 such events and has one, and the calibration sample produced its twin independently (Table 7). The fourth stage-1 outlier, towards CP–72 2713 in Band 7, never acquired an astrophysical attribution and was marginal throughout, its $T_\star = 5.81$ standing against a ring maximum of 5.68. Its prior for an astrophysical line is high: a K7/M0 member of the ~24 Myr β Pictoris moving group with a cold, dust-rich debris disc (Moór et al. 2020), the same population as AU Mic. It was recorded as an unattributed stage-1 outlier under criteria pinned in advance (*promote* on independent confirmation, *retire* on a clean second epoch or an identification), committed on 2026 September 11 (119a68a6ab50), 9 hours before the second block was searched.

That block, A002_Xff0235_X502d, sits in the same member observing unit set and was declined by the de-duplication of §3; we have since calibrated and searched it with the unmodified pipeline. It matches the first in tuning, channel width, channel count, integration count and on-source time, with frequency axes registered to 0.18 of a channel, and is the deeper of the two (1.964 against 2.097 mJy), so a persistent emitter at the first block’s flux would have appeared at $T_\star = 6.21$, above both the trigger and that window’s largest control 5.93. The margin is narrow, so a second stage-1 outlier was never assured even for a persistent emitter, and the disposition rests on the flux comparison, which carries no

trials factor.

The feature does not recur. The first-block feature drifts. Its maximum lies at a drift of -2.06 kHz s^{−1}, a track sweeping 15.8 channels across the block, so the test belongs at that frequency *and* that drift. There the repeat block measures -1.0 ± 1.8 mJy, -0.6σ , against 11.3 ± 2.0 mJy in the first. Read as two measurements of one constant amplitude, the pair is inconsistent at 4.6σ ; with the first block’s flux taken as exact, which flatters the test because that flux is the largest value in the window and so biased high, the repeat block alone excludes it at 6.8σ . Maximised afresh over the repeat block’s own drift grid, the same channel returns $T_\star = 3.60$ against 5.81, worth 1.6σ with an arbitrary recovered rate, and that weaker comparison is the form in which such tests are usually quoted. Two executions cannot separate a noise excursion from an intermittent emitter, and we do not claim they do.

No solar-system body was in the field. A SkyBoT cone search of radius 30 arcmin about the pointing at the block’s epoch returns 0 known minor planets; the field lies 56° from the ecliptic and the nearest major body, Saturn, is 57° away. The same query on the ecliptic at the same epoch returns 118–444 objects, so the null return measures the field rather than a failed query. An *uncatalogued* body is not excluded by that search, but it is excluded by the geometry: a solar-system object moves non-siderially, so it could not hold the same position and the same topocentric frequency in two blocks 2.05 h apart at one phase centre, and the feature is absent from the second block in any case. Resolving the remaining ambiguity needs a third epoch, and we intend to request one: a single Band 7 execution reproducing the tuning would separate an intermittent emitter from a noise excursion, which two blocks two hours apart cannot.

One qualification travels with this result: the two blocks are a single night’s pair, 2022 October 2, the second beginning 2.05 h after the first. That makes the test strong against instrumental classes the scramble null cannot reach, a correlator spur or fixed-channel band-pass residual being at its most reproducible two hours later in the same configuration, and none reproduced; and weak against intermittency, since it constrains persistence across two hours alone and says nothing about longer duty cycles.

Four further diagnostics – the neighbouring-channel

and window-maximum statistics of the repeat block, the local scramble test, the edge-channel and ring-margin details, and the debris-belt, split-half and continuum checks – are in Appendix G. None changes the disposition.

Frequency disposes of the catalogue entries. A catalogued transition is stationary in the observed frame to well within a channel over two hours, while at zero drift the first-block statistic is only 2.02σ . Re-querying the wider catalogues at the corrected 344.269746 GHz returns 3 entries in the $\pm 50 \text{ km s}^{-1}$ tube, but such proximity carries no discriminating power: the same query at the 20 crossing frequencies returns a transition within 60 km s^{-1} for 19 of them.

Polarisation hands, baseline splits and calibrator comparison still need re-extraction beneath the frozen products, but the visibilities themselves have now been tested. On an independent recalibration of the first block the drifting channel was fitted against no source, a point source at the star and a point source anywhere in the beam, with the local continuum removed so the star’s own 10.9 mJy is not read as a narrowband excess. Over 405 860 visibilities the fit returns 10.4 mJy at the star, 5.5σ against 1.89 mJy noise ($\Delta\chi^2 = 30$), an imaginary part of 0.03σ as a real source on the phase centre requires, and a best-fitting position on the star. At 4 control frequencies the statistic at the star never exceeds 1.7σ while the maximum over 961 trial positions reaches 4.4–5.4 σ , so a grid maximum of this size is unremarkable but one on the star is not. The feature is point-source-like and at the stellar position in the first block, a stronger statement than the image-plane statistic alone supported, and the disposition is unchanged: the event is unconfirmed and single-epoch, with persistent emission of the same strength excluded on the 2.05-h baseline. Persistence is the survey’s operational criterion, not a physical requirement on a transmitter: an intermittent emitter is not excluded by two executions two hours apart, and we do not claim otherwise.

The pinned criterion is met on the matched-frequency, matched-drift null of the repeat block. The standing description of this window is therefore an unconfirmed first-block threshold event that is absent in the repeat block. We find no repeat and so no evidence for a *persistent* technosignature at this frequency; the event itself remains unexplained, is statistically unexceptional at the survey level, and its absence two hours later does not identify it as noise. Two executions do not exclude an intermittent emitter, and we do not say they do.

A non-recurrence means nothing unless the same machinery recovers a repeat when one exists, so we applied it to emission we already believe in. The unit behind the Band 3 flag holds 4 blocks, all 4 now searched unmodified, an exhaustive control. The CO(1 $\rightarrow$ 0) feature is present in every one at $T_\star = 14.65, 17.29, 27.68, 30.76$, with 17, 24, 24, 24 channels above threshold and the control maximum below the star throughout. Those executions share one scheduling block within 23 h, so they are one epoch repeated. A second Band 6 CO(2 $\rightarrow$ 1) window at 230.528134 GHz repeats the test in another exhaustive unit, 2 of 2 blocks searched. It crosses the threshold in both without becoming a stage-1 spatial outlier, at $T_\star = 10.16, 9.95$ on an exact configuration match. The integrated line fluxes agree to -0.7σ . It is Class A, so the recurrence is demonstrated at the survey’s

finest drift discrimination. The stage-1 Band 6 window at 230.516128 GHz has since been tested on the same terms. Its unit’s second block, A002_Xd9668b_Xa9df, was calibrated and searched with the unmodified frozen pipeline on 2026 September 12 in the configuration of the first, 385 integrations of 3,534 channels at 15.26 kHz, and the feature recurs. The star reaches $T_\star = 8.95$ over 106 channels above threshold, 102.5 kHz from the first block’s peak, and that separation is 0.13 km s^{-1} against a line 8.9–10.2 km s^{-1} wide. Every β Pictoris CO feature this survey flagged has now been recovered in an independent execution block, and CP–72 2713’s has not.

The spatial ranking carries a warning with it, and we state it plainly. The repeat block is again a stage-1 outlier, none of the 512 controls reaching the star, but the margin has collapsed, from $1.28\times$ its largest control to $1.06\times$ as the star falls from 11.68 to 8.95. The controls carry the diagnosis: their maxima, 9.12 and 8.41, stand against 99th percentiles of 6.11 and 5.99 in the two blocks, so in each a few control positions hold the belt while the rest hold noise. Pinned to the first block’s own peak frequency instead of maximised over the window, the star returns 8.10, and 2 controls exceed that. The cause is the one already identified for the neighbouring Band 6 window: the CO belt fills the control annulus, so the controls rise with the star. The spatial screen loses power against emission extended on the scale of the annulus. That is a property of the method rather than of this line, and it is why a recurrence test at a known frequency is the stronger instrument.

The recurring window also locates the emission. In both blocks the annulus is brighter than the star (ring maxima 15.17, 15.08 against $T_\star = 10.16, 9.95$, 8 and 6 of the 512 controls above the star), and that annulus, 3.82–21.27 arcsec, sits on this disc’s CO belt, peaking near 85 au = 4.3 arcsec at 19.6 pc (Matrà et al. 2017). The window is therefore not a stage-1 outlier because the source fills the ring, and the frozen products place it off the star, though only coarsely: the window’s 4.29-arcsec synthesised beam exceeds r_{in} , so the innermost controls are not resolved from it.

Frequencies are compared after removing each block’s own tuning, which ALMA sets per date, absorbing most of the barycentric difference; on that basis the Band 3 peaks agree to 1.00 channel and the Band 6 peaks to 3.00 channels. In the barycentric frame, and now in the ordinary radial-velocity convention rather than the frequency-offset sign of Table 15, the CO peaks of the 4 blocks with a recorded epoch span +19.6, +20.2, +20.4, +20.7 km s^{-1} , within 0.7 km s^{-1} of the adopted systemic velocity across two transitions and 8.4 yr, 2013-10-06 to 2022-03-03. Widths place the same conclusion beyond a frequency coincidence: the channel counts above threshold correspond to 10.8–15.2 km s^{-1} in Band 3 and 8.9–10.2 km s^{-1} in Band 6, both an order of magnitude above the one channel a carrier occupies. The β Pictoris test establishes that the machinery preserves a real repeat; what establishes that *this* repeat would have been seen is the $T_\star = 6.21$ expectation above.

5.3.10. Line exclusion: definition and cost

The exclusion regions are defined before any crossing is classified, and they are *excluded parameter space*, not a veto: the experiment declines to classify those frequen-

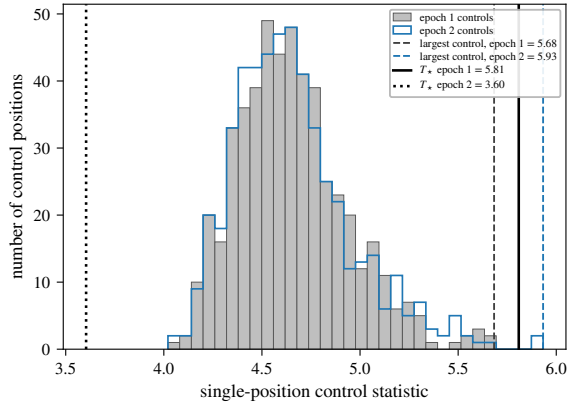


FIG. 5.— The CP-72 2713 Band 7 crossing inside its own control ensemble, in both executions. Histograms give the 512 single-position control statistics of that window in execution 1 (filled) and in the second block 2.05 h later (outline); the two agree, which is the evidence that the datasets are comparable. Vertical lines mark T_* and the largest control in each. In execution 2 the control annulus alone reaches 5.93, beyond the 5.81 that raised the flag, while the star falls to 3.60.

cies rather than declaring anything found in them astrophysical. The searches as executed matched crossings to catalogued rest frequencies within one spectral channel at zero drift. The β Pictoris windows showed that inadequate: both fell outside that tolerance, by 10.6 and 21.9 MHz, while lying within 3.6 km s^{-1} of the stellar systemic of a resolved, mapped disc. Every crossing was therefore reclassified with a velocity-space mask, $\pm 50 \text{ km s}^{-1}$ in the stellar frame around each catalogued transition, whose kinematic justification and the counter-argument that a communicator might choose these natural reference frequencies are in Appendix D. It is a post-hoc classification criterion and never a detection threshold, and its half-width was adopted after the β Pictoris crossings had been seen. Reclassifying all 20 crossings at ± 20 to $\pm 100 \text{ km s}^{-1}$ costs 2.1 to 10 per cent of gross bandwidth, admits no new unattributed outlier, and leaves CP-72 2713 the only unmasked one at every width. Masked regions are reported throughout as *excluded search space*, not as searched-and-empty. What the mask cost in candidates, as opposed to bandwidth, is nothing: 7 of the 20 crossings fall inside a tube, 3 of them are the stage-1 outliers already dispositioned as CO in Table 6, and the remaining 4 do not exceed their own control ensembles, so no masked crossing would have entered candidate generation had the mask been absent.

The mask applied holds 17 transitions of 11 species at full JPL/CDMS precision (Pickett et al. 1998; Müller et al. 2005), applied in the LSR frame as well as the stellar frame and re-applied to the frozen crossing list as a post-hoc filter. The species criterion is the one objective rule we could state in advance: the transitions ALMA’s own band definitions are built around, namely the CO isotopologues, HCN, HCO^+ , CS, SiO, SO, CN, H_2CO and [C], which dominate the circumstellar and foreground emission of disc-selected fields. The choice is consequential rather than incidental: a full Splatalogue selection would mask 75.4 per cent of the union against 0.78 per cent here, and would exclude 19 of 20 crossings (Appendix D), so “coincident with a known molecular line” is not a binary property but a statement about a catalogue. Since no masked crossing exceeds its own control ensemble

(above), no plausible widening or narrowing of the species list changes a disposition. The cost of masking is measured on one definition, every JPL catalogued transition of the 9 frozen species displaced by each system’s radial velocity, given a $\pm 50 \text{ km s}^{-1}$ stellar-frame half-width, intersected with each window and merged: 3.9 per cent of the unique-frequency union and 4.4 per cent of the window-summed gross bandwidth, with CN and CO dominating the loss. Total exposure is $1.5 \times 10^6 \text{ s GHz}$, or 393 star-hour-GHz, with 30 per cent of it in the five best-populated 2-GHz bins, which any frequency-prior calculation must weight (Appendix C). The mask governs dispositions and leaves the noise-derived thresholds alone. Table 12 gives the accounting band by band rather than as one survey total, because the loss is not uniform: it reaches 6.5 per cent in Band 9 and stays below 1.5 per cent in Bands 3–5, so a per-band frequency prior cannot use the global figure.

Masking is a design selection effect, and the distinction that matters is between *discovery* and *classification*. For discovery, line frequencies need not be removed at all: a carrier there is as detectable as anywhere else. For classification they are decisive, because the astrophysical prior at a catalogued transition is overwhelming and this pipeline cannot separate the two hypotheses on morphology alone. Removing them from the searched domain is the conservative choice and the one a non-detection can be quoted against; the better design, which we recommend for a successor, searches every frequency and attaches a *lower prior* to a line-coincident event rather than deleting the channels. A signal coincident with a molecular line is not thereby “false”, and these are exactly the frequencies a communicator wanting to be found near a natural reference might choose. We place no constraint on transmitters in the excluded regions, 3.9 per cent of the unique-frequency union, and the release keeps those channels for the second of the two searches this dataset supports: a clean-frequency search, in which line coincidence is a veto, and a line-coincident search, in which a crossing is admitted only on independent evidence of spatial localisation, line morphology, polarisation, recurrence or non-thermal width. This paper reports the first; the β Pictoris crossings are what the second would have to handle. The exposure is not confined to the flagged windows either: 131 of 443 windows contain an in-band catalogued transition of the masked species, and in none of them does a crossing fall within one native channel of a rest frequency.

5.4. Audit against a region-maximum statistic

The statistic of Eq. 2 evaluates star and controls identically. A region maximum taken over $n_{\text{src}} = 135$ positions at the star, ranked against single-position controls, is not exchangeable with them and ranks first with probability $n_{\text{src}}/(n_{\text{src}} + 512) = 20.9$ per cent rather than $1/(512 + 1)$; that is a property of the estimator class, derivable without reference to any window’s outcome. The primary analysis uses the symmetric form throughout and all 443 windows are scored under both. Table 9 is the audit: the symmetric criterion flags a strict *subset*, seven windows against four, the three that drop out doing so because the stellar position never exceeded its own control ensemble. Both columns are released for every window, so the dependence of the result on the choice can be checked di-

rectly. During the analysis the symmetric form replaced the region maximum, 9 days after the region maximum had flagged a crossing towards AU Mic which the symmetric form does not flag; we state that plainly and let Table 9 carry the audit.

61 stars hold further public blocks this survey did not take (§3), which §6 turns into an epoch-activity constraint.

5.5. Ancillary results

Three by-products carry no technosignature disposition: an integration-time audit (two of 58 star-execution-block groups over-count on-source time, the rest conservative, and no EIRP threshold depends on it); the continuum lane’s 57 target/bands, seventeen with $\geq 5\sigma$ detections, all astrophysically unsurprising bar the auroral emitter LSR J1835+3259 (Appendix E); and a serendipitous line catalogue of twelve outlier groups, six plausibly real, released with the catalogue.

6. DISCUSSION

6.1. What signals could ALMA archival data actually detect?

Each morphology is illustrated with a real window, the completeness entries of Table 4 and Fig. 6. Denominators count the 81 systems whose deepest threshold in the class lies at or below the case power.

(i) *A continuous, frequency-stationary carrier* in a coarse channel is the best case. The UV Ceti pair at 2.7 pc has a deepest window whose trigger is 1.6×10^{13} W and whose effective threshold is 3.7×10^{13} W, and a channel-confined carrier there is recovered at near-ideal $\sqrt{N}$ gain (§4.5). Read on the trigger, 1 system of 81 reaches the numerical EIRP of the Arecibo planetary radar and 2 reach twice it; read on the effective threshold, which is the comparison a transmitter power calls for, 0 do and 1 does. The survey’s best case is therefore a few times that numerical EIRP.

(ii) *A continuously drifting carrier* in a fine 488-kHz channel is the only morphology with a measured end-to-end recovery curve (Appendix B): above $\sim 8 \times 10^{13}$ W nominal it is recovered in 83 per cent of trials with its drift inside the searched ceiling, 53–94 per cent once the $0.5\text{--}2\times$ transfer bracket is carried, and 8 systems reach that depth in Class A.

(iii) *An intermittent carrier, on for a quarter of the track*, above $\sim 2 \times 10^{14}$ W is detected toward 24 mid-distance systems. Recovery falls rapidly at smaller dwell fractions (Table 4): below $f_{\text{dwell}} \sim 0.1$ only the nearest windows carry information.

6.2. What does the null result mean?

No credible technosignature has been found (§5.3). The $\text{EIRP}_{5\sigma}$ values are nominal trigger powers (§4), and completeness-calibrated exclusion holds only in Class A, where the measured recovery curve applies.

One exclusion deserves the converse emphasis. The line mask removes $\pm 50 \text{ km s}^{-1}$ stellar-frame tubes around the catalogued transitions, whose rest frequencies are the ‘magic’ channels Cocconi & Morrison (1959) argued a communicator might choose. This survey is blind by construction to a transmitter parked there in the stellar frame, a false-positive control purchased with 4.4 per

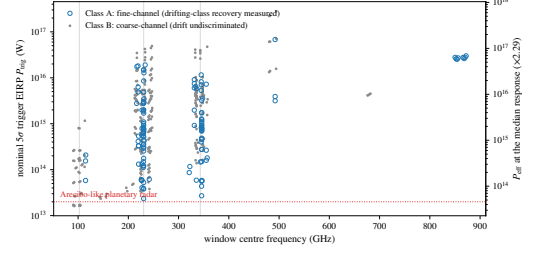


FIG. 6.— Two-dimensional sensitivity against window centre frequency for all 443 searched windows, symbol by channelisation class, the axis along which completeness differs (Table 4). *Left axis*: the nominal trigger threshold P_{trig} , an internal detection statistic. *Right axis*: the effective threshold at the *median* response, $P_{\text{eff}} = 2.29 P_{\text{trig}}$, which is the power to compare with a transmitter (§4); the released per-window ratio $P_{\text{eff, total}}/P_{\text{trig}}$ runs 1.33–3.98, so read a single window’s threshold from the catalogue and not from this axis. Ringed symbols are the 9 windows whose intra-integration smearing correction exceeds 1 per cent. Grey lines mark the Bands 3, 6 and 7 concentrations; dotted, the Arecibo 2.38-GHz EIRP benchmark, a power scale only. Note the two decades of scatter at fixed frequency: there is no single survey sensitivity.

cent of gross bandwidth and one physically motivated class of signal. A carrier deliberately placed on CO(2→1) is not undetectable by ALMA, it is unsearched here, and those channels are retained in the release for anyone bringing independent spatial or time-domain discrimination. Of the 41 planets known around the 18 hosts, only TRAPPIST-1 b reaches the drift ceiling in its edge-on worst case, on accelerations $GM_{\star}/r^2$ from NASA Exoplanet Archive parameters maximised over phase and inclination. Figure 8 shows the ceiling as a selection function of period and host mass, the host-mass dependence being weak ($a \propto M_{\star}^{1/3}$ at fixed period). Coverage eligibility and search completeness differ: §3 audits the 451 public blocks in scope against the 104 searched, and the campaign of §5.3.1 is working through the remainder.

The survey sits against Wright et al. (2018)’s haystack axis by axis. On frequency only Band 3 falls inside that axis’s 10 MHz–115 GHz span, 20.6 of 115 GHz, and the remaining 93.2 GHz (Bands 4–8) lies wholly outside it and is new territory. On sensitivity the median threshold detects a 10^{13} W transmitter only within ~ 0.9 pc. Against the sole prior ALMA-archival search (Mason et al. 2024, 28 bycatch targets at ≥ 1.01 kpc), the 81 systems here reach median per-target thresholds $\sim 1250\times$ deeper. We do not combine these fractions into one coverage number, because the axes are strongly coupled and a product of marginals would misstate the overlap by orders of magnitude.

We do not convert the zero detection into a fraction of systems hosting transmitters. Doing so requires a per-system detection probability conditioned on this survey’s frequencies, epochs, duty cycles and morphology class, on a sample selected by ALMA’s own proposal history, and on a transmitter-frequency prior confined to the searched islands; under any broader frequency prior the bound vanishes, since the per-system Class A union is 1.73 GHz in the median against ALMA’s ~ 75 GHz of tuning range. The result of this paper is the searched frequency–time–power volume, the measured recovery within it, and the null.

How far that null reaches is set by recurrence, not by the rank statistic. No single event in this design can clear the survey-wide threshold (Appendix F), so a genuine

TABLE 9

EVERY WINDOW FLAGGED BY THE ORIGINAL REGION-MAX STATISTIC, UNDER BOTH STATISTICS, SO THE REVISION CAN BE AUDITED ACROSS ALL OF THEM, NOT JUST THE AU MIC WINDOW. S_{reg} : REGION MAXIMUM OF EQ. 2 OVER $n_{\text{src}} = 135$ POSITIONS; T_* : THE SYMMETRIC SINGLE-POSITION STATISTIC; ‘CTRL MAX’, ‘RING MAX’: THE LARGEST CONTROL STATISTIC UNDER EACH. THE ν COLUMN IS EACH WINDOW’S CENTRE; THE CROSSING FREQUENCY DIFFERS BY UP TO 0.85 GHz, AND TABLE 6 CARRIES ν_{cross} . FOR CP–72 2713 THE REGION MAXIMUM ALREADY LAY ON THE STAR, SO THE TWO COINCIDE.

Star	Band	ν (GHz)	region-max statistic		symmetric statistic		Symmetric outcome
			S_{reg}	ctrl max	T_*	ring max	
HD 48370	6	230.7305	27.95	26.83	27.10	26.83	flagged
β Pic	3	115.2605	20.31	7.27	14.65	7.27	flagged
β Pic	6	230.5164	12.78	9.12	11.68	9.12	flagged
AU Mic	6	230.5384	5.97	5.85	5.22	5.85	not flagged
ALMA J1537–3319	6	217.0183	5.83	5.63	5.09	5.63	not flagged
CP–72 2713	7	345.1152	5.81	5.68	5.81	5.68	flagged
HD 14055	7	345.1377	5.80	5.74	5.22	5.74	not flagged

but weak transmitter seen in one epoch is not separable from the 1.2 chance events the survey expects, which is exactly the position CP–72 2713 leaves us in. What separates them is a second epoch or an independent astrophysical identification, and only 18 of the 81 systems, 22 per cent, were searched in more than one execution block. A further 45 hold archived blocks that were not both processed, 63 in all, so the recurrence test could in principle be extended to 78 per cent of the sample with no new observing time.

In plain terms, ALMA has observed 88 nearby stars as a by-product of other science, across parts of the millimetre band no technosignature search had examined, and we found no channel-confined carrier the surrounding sky could not also produce. On the few gigahertz, few hours and single morphology class each star happened to be observed with, that says little about how common transmitters are.

6.3. What this sample can and cannot speak for

Selection is on archival coverage, and ALMA’s archive was not built to sample stars. 79 of the 88 searched stars were observed under proposals categorised *Disks and planet formation*, and 73 carry the *Debris disks* science keyword (67 of 81 systems). Among the stars with a Gaia temperature, 41 per cent are M dwarfs against 69 per cent of the reference census (Table 10), so the sample is F- and G-enriched and M-dwarf deficient. Window counts are concentrated too: three systems carry 9.0 per cent of the searched windows.

Two consequences follow, and they run in opposite directions. A debris-disc-dominated sample is the worst case for astrophysical false positives at millimetre wavelengths, which is where the stage-1 outliers of §5.3 concentrate, so the vetting burden here is heavier than a blind sample would carry. And because M dwarfs host most of the habitable-zone planets in the solar neighbourhood, **the under-representation of M dwarfs means this result carries very little demographic weight for the local habitable-zone planet population**: only 21 of the 5908 M dwarfs within 40 pc carry a searched window, 0.4 per cent, both counts being restricted to the temperature-classified sample of Table 10 and therefore illustrative of the selection rather than a complete selection function. Conversely, the over-representation of F and G stars means Sun-like stars are sampled relatively heavily, though still through a highly non-random, disc-selected archive. Either way it constrains transmitters around the stars ALMA happened

to point at, not the local stellar population.

Useful subsamples still fall out of archival selection. 18 of the 88 processed stars (41 known planets) are confirmed exoplanet hosts, among them Proxima Centauri, τ Ceti, ϵ Eridani, β Pictoris and the seven-planet TRAPPIST-1 system, while HN Lib and LHS 1140, each with a habitable-zone planet, now carry an ALMA technosignature non-detection. Four searched stars are white dwarfs (Gentile Fusillo et al. 2021), all continuum non-detections and among the few mm/submm SETI constraints for that class (Huang et al. 2026).

6.4. A next-generation ALMA technosignature search

ALMA’s advantages here are its mm-wave sensitivity, an essentially unsearched tuning range, the interferometric spatial discrimination that supplies the control statistic of §4.2, and a public archive re-processable without new allocations. Six changes would enlarge what this experiment can say, ranked by the signal space each buys.

Search the blocks already held. The public blocks outside this analysis (§3) cost no telescope time, and analysing them would make a multi-epoch survey of 61 stars. A survey that keeps growing has no fixed denominator for its false-alarm accounting, which is why the analysis reported here has one. One caution carries over from the worked case, whose two blocks are repeat executions of a single scheduling block on one night: a second block often supplies a baseline of only hours, and which it is can be settled with one archive query before any download.

Use the visibilities. This is the principal methodological lesson of the survey, and §5.3.2 is why. A celestial point source at the stellar direction $\mathbf{x}_*$ contributes

$$V_{ij}(\nu, t) = A(\nu, t) \exp(-2\pi i \mathbf{u}_{ij} \cdot \mathbf{x}_*), \quad (6)$$

so rotating the phase centre to the star should leave a coherent signal entirely in the real part, with an imaginary component consistent with zero on every baseline. That test asks whether the data contain a point source *where the star is*; the control ranking asks only whether one sky extraction is unusually large compared with neighbouring ones, which is a weaker question and, as §5.3.2 shows, one whose null depends on where in the field the comparison positions sit. Applied to the one unattributed window (§5.3) the visibility fit was decisive; applied to every window it would demote the control ensemble to a calibration layer, remove the radius-matching problem entirely, and give each candidate a physically specific test rather than a rank. Doing so needs the visibilities re-

tained, which this survey did not do, and it is the first change we would make.

Drift discrimination on coarse windows. Within the searched ceiling every in-grid drift on a 15.625-MHz window is sub-channel over a track, so 317 of 443 windows measure amplitude but cannot distinguish a drifting carrier from a stationary one (§4.5). Finer channelisation, or a drift model spanning several blocks, would recover that axis.

Retain more control spectra. Full dynamic spectra survive for only 8 of the 512 control positions, which is what breaks the second-stage null in a structured field and what limits the variance test of §5.3.5. Retaining all of them costs storage and nothing else.

Extend the injection campaign and the polarisation axis. The stratified campaign of Appendix B now spans band, channel width, integration time and array, and what remains unspanned is other target environments and edge-phased drift grids. The polarisation axis buys discrimination against weakly polarised astrophysical backgrounds at no sensitivity cost.

A boundary on drift linearity. The search assumes linear drift over a track, and curvature matters once $\frac{1}{2}\ddot{\nu}T^2$ exceeds half a native channel, which only the shortest-period orbits reach: $P \lesssim 1.6$ d for a typical Class A window (230 GHz, 488 kHz, $T \approx 2000$ s). Among the 41 known planets of the 18 hosts in the sample, 1 lies inside that boundary: TRAPPIST-1 b at 1.51 d, which is also the case that reaches the drift ceiling. The next shortest period is TRAPPIST-1 c at 2.42 d and the shortest outside that system is GJ 581 e at 3.15 d, both comfortably linear; periods are catalogued for 39 of the 41, the exceptions being directly imaged companions on orbits of decades, which are further from the boundary still.

Budget for the disc-CO false positives, but do not pre-judge the targets. 3 of the 4 stage-1 outliers are molecular emission toward disc hosts, arising in 2 of the 67 debris-disc systems searched: one flagged system in 34, or 2.4 per cent of Class A windows. That is the natural false-positive rate an archive proposed for disc science carries, and any future search on ALMA, NOEMA or the JVLA inherits it. The remedy is procedural: run the continuum lane (Appendix E) *before* promoting a window, which would have caught HD 48370 and both β Pictoris windows at the first gate. We do *not* recommend lowering the threshold toward disc hosts: a detection statistic should not know whether its target is astrophysically interesting, or a real transmitter would be penalised for the company it keeps. Disc membership belongs in classification, where it already is.

7. CONCLUSIONS

What was searched. We searched a defined subset of public ALMA observations covering 88 stars in 81 systems within 40 pc: 109 target/band datasets and 443 spectral windows providing 113.9 GHz of *gross* unique frequency coverage between 89.6 and 873.1 GHz, of which 109.4 GHz remains as the *effective unmasked* technosignature-search union after the molecular-line exclusion. The sample is complete with respect to the frozen candidate work list rather than the volume, and archive-incomplete: 104 of 451 public execution blocks were analysed, and the searched stars are 0.50 per cent of the catalogued population within 40 pc. Channelisa-

tion splits the search in two: 126 windows of fine-channel carrier search against 317 of unresolved spectral excess (§4.2).

What was found. Within the 126 fine-channel windows toward 59 systems and the 317 coarse spectral-excess windows toward the remainder, of which 132 resolve drift on the physical criterion, we find no persistent, independently confirmed spectral technosignature at the calibrated sensitivity of the archival observations. The $\pm 50 \text{ km s}^{-1}$ molecular-line mask is part of that qualification: it removes 4.4 per cent of gross bandwidth around transitions whose rest frequencies are exactly the channels a communicator might choose, and those channels are unsearched here because the analysis vetoes them, while ALMA can see them perfectly well. Four windows are stage-1 spatial outliers, and three of those are CO emission: two circumstellar components towards β Pictoris and one previously reported foreground cloud towards HD 48370 (Cataldi et al. 2023). The fourth, towards CP-72 2713, never acquired an astrophysical attribution. A second public execution block at the same tuning, searched after the criterion had been prospectively frozen in the released repository, shows no repeat at the feature's own frequency and drift. We therefore find no evidence supporting a *persistent* technosignature, and report 0 credible ones; two executions cannot separate a noise excursion from an intermittent emitter (§5.3). Barnard's Star and Wolf 359 carry their first published mm/submm limits (§5.1).

The defensible sensitivity and search domain. The nominal EIRP_{5 σ} values are trigger powers, not completeness limits (§4): 2.3×10^{13} – 6.8×10^{16} W over Class A and 1.6×10^{13} – 2.8×10^{17} W over Class B. The number to set against a transmitter is the response-corrected threshold, 5.4×10^{13} – 1.6×10^{17} and 3.7×10^{13} – 6.4×10^{17} W, on which 0 of the 81 systems reach the numerical EIRP of the Arecibo planetary radar. Neither is a completeness limit: recovery is calibrated end to end on one configuration and transfers within 0.64 – $1.13\times$ over 28 more (Appendix B). The *Scope of the constraints* box of §4 fixes what the non-detection binds. We quote no transmitter occurrence fraction: any such number conditions on the searched islands, a duty cycle, one morphology and a target list ALMA chose for disc science, so the only defensible cross-survey comparison is between EIRP thresholds themselves (Fig. 1).

What the result is, and is not. The control ensemble is an empirically calibrated screen for extreme values, not an exactly exchangeable rank test: out of sample the star's rank is displaced (median 0.44 against 0.5), control positions containing no star are displaced as far, and the remedy is geometric, a visibility-domain test of whether a point source sits at the stellar position (Eq. 6). In the tail on which every disposition depends the screen is calibrated to a factor 1.4, so the false-alarm rate is quoted from measurement, and the zero-candidate result is unchanged under either form of the statistic (Table 9). No persistent, channel-confined spectral technosignature was identified in the searched frequencies and archival epochs. That constraint is strongly conditioned on channelisation, on sparse frequency coverage, on a target list ALMA chose for disc science and on transmitter duty cycle, and it is not a demographic statement

about transmitters around nearby stars. The result that does generalise is methodological: ALMA archival data can be made to yield calibrated mm and sub-mm technosignature limits, with measured completeness and a measured false-positive rate, and the limits are set by archival channelisation, heterogeneous depth, line confusion in a disc-selected target list and sparse temporal sampling rather than by sensitivity. 347 of the 451 public execution blocks in scope remain unanalysed and need no new observing time; a successor should retain visibilities, match controls in radius, prefer finer channelisation, revisit epochs and keep both polarisation hands (§6.4).

ACKNOWLEDGEMENTS

This paper uses ALMA data from the public ALMA Science Archive. The 104 execution blocks searched span 2013 October 6 to 2025 June 11 across 36 proposal codes; their project codes and epochs are listed in the machine-readable release and should be cited from it in the standard JAO.ALMA#XXXX.X.NNNNN.S form. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), MOST and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia*, processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database (CDS, Strasbourg, France).

DATA AVAILABILITY

The reduced data products underlying this paper, with the full target-selection and analysis pipeline, are at <https://github.com/Tilanthi/SETI>. All numerical values here come from the *submitted-analysis snapshot*, the repository commit tagged at submission, which freezes pipeline code and input data as run and reproduces every number, table and figure from the regeneration scripts shipped with it.

The machine-readable catalogue, `per_target_results_v3.68.csv`, carries 443 rows, one per searched window, in 45 self-described columns: identification and distance, band, execution block and searched frequency range, channel width, exposure and per-channel rms, the nominal EIRP_{5 σ} trigger with both response-corrected values (§4), the drift ceiling, trial count, η_{drift} and η_{smear} , resolution and search class, the stellar and control peak statistics with the add-one rank, the crossing frequency and nearest catalogued transition, the full control geometry including its seed, and the disposition verbatim from Table 6. The statistical unit is `system_id`, with seven bound pairs sharing one identifier. Deleting every generated table and figure and rebuilding from the scripts alone reproduces all 49 products byte for byte. Blocks searched by the epoch-extension campaign of §5.3.1 lie outside the frozen release and ship in the accompanying epoch-extension record. The release also builds diagnostic figures the paper does not print, among them the per-band noise-quality plot and the illustrative occurrence curve; they are products of the same generators and carry no number quoted here.

The version of record is a frozen archive, not a branch:

the Zenodo deposit carries the exact commit that produced every number, table and figure in this paper, the 443-row catalogue, the control statistics, the injection records and the calibration sample as a separately labelled product, and the repository tag resolving to that commit is named in the deposit. A reader who regenerates from the deposit reproduces this paper; the GitHub branch may move on. The deposit, with a DOI minted on acceptance and inserted here, carries the per-window search products for all 443 windows, each with its 512 control statistics, so every rank and p -value here is recomputable; the line mask in frozen and repaired forms; the execution-block metadata and pointing ledger; the 1200 injection, 3888 dwell and 4032 stratified trials; the local-scramble code and outputs; the repeat-block and visibility products behind the CP-72 2713 disposition; and the frozen pipeline and data snapshot with the change-log in which the promote and retire criteria are frozen in the repository before the data are seen.

Data products carry the Creative Commons Attribution 4.0 International licence (CC BY 4.0) and the pipeline code the MIT licence. Raw ALMA visibilities are public from the ALMA Science Archive at <https://almascience.org>.

SOFTWARE

The pipeline depends on CASA 6.7.6 (casatools/casatasks 6.7.6-14; CASA Team et al. 2022), Python 3 with astropy (Astropy Collaboration et al. 2022), numpy (Harris et al. 2020) and scipy (Virtanen et al. 2020), and the ALMA Science Archive and SIMBAD TAP interfaces through astroquery (Ginsburg et al. 2019); all are free and open-source, with exact versions recorded in the frozen release.

ABOUT THIS PAPER

This paper was built using the Open Journal of Astrophysics L^AT_EX template. The OJA is a journal which provides fast and easy peer review for new papers in the `astro-ph` section of the arXiv, making the reviewing process simpler for authors and referees alike. Learn more at <http://astro.theoj.org>.

APPENDIX

A. FREQUENCY CONVENTIONS, LINEWIDTH READING, AND SMEARING

Reference frames. The line mask’s chain runs topocentric $\rightarrow$ barycentric (Earth ephemeris at the execution-block mid-time) $\rightarrow$ stellar rest frame (catalogued systemic radial velocity), with JPL/CDMS rest frequencies (Pickett et al. 1998; Müller et al. 2005); in the LSR frame it stops at the standard solar motion, not the systemic velocity (§5.3.10). The residual error is the catalogue radial-velocity uncertainty alone, a few km s^{-1} within 40 pc; second-order relativistic terms at the 50 km s^{-1} scale are $\lesssim 5 \text{ kHz}$ at 350 GHz, below the finest native channel searched (15.3 kHz). The β Pictoris and AU Mic attributions use the same chain, so their offsets are comparable to the exclusion tolerance.

Channel widths, and the instrumental response. These come from the measurement sets’ own spectral window tables (CHAN_WIDTH). Earlier versions read a missing “EFFECTIVE_BANDWIDTH” column as evidence

of no online spectral averaging; MS v2 defines no such column, so that check is withdrawn, and with it the claim that the quoted channelisations are noise bandwidths.

ALMA applies online Hanning smoothing by default, so the spectral response is ~ 2 channels wide and neighbouring channels are correlated. For the 0.25/0.5/0.25 kernel the coefficients are $\rho_1 = 2/3 = 0.667$ and $\rho_2 = 0.167$, with $\sigma_{\text{smoothed}}/\sigma_{\text{raw}} = 0.612$: the correlation is not confined to adjacent channels, the lag-two term being a sixth. A sub-channel tone keeps 0.50 of its power in the peak channel at the channel centre and 0.38 on a boundary, median 0.44 over uniform placement. Every $\text{EIRP}_{5\sigma}$ here is therefore optimistic by $\times 2.00$ to $\times 2.67$, median $\times 2.29$; the released catalogue carries the nominal threshold and both corrected values per window, and the sensitivity summary quotes the worst case. The injections deposit an unsmoothed delta-in-channel tone, so the recovery curves of §4.5 and Appendix B carry the same bias and the end-to-end check (Eq. 5) cannot see it. The retained per-integration residuals of the second-stage local null (§5) have a measured lag-one spectral autocorrelation of 0.65, against the Hanning prediction 0.667, so that correlation is instrumental and the residual spectra are smoothed and un-averaged, which is the independent confirmation of the smoothing state. The correlation model is truncated at lag two; the baseline step is a per-integration median in blocks of up to 65 channels, which could imprint correlation on longer scales, and the released products retain no residual spectra with which to measure it. If such a term is present the independent-trials count of Appendix F is an over-estimate and the achievable per-window resolution is coarser still, which strengthens rather than weakens the conclusion that rank alone cannot authenticate a candidate. The archive’s reported effective spectral resolution per window, against each measurement set’s own channel separation, is exactly 2.00 in 403 of the 431 windows, in both correlator modes and including 309 of the 313 coarse time-division-mode (TDM) ones; the other 28 report a smaller ratio, the signature of online channel averaging, where the peak-channel loss is smaller, and the 12 Band 9 and 10 windows, absent from the metadata snapshot, take ALMA’s default. The blanket factor is right for the great majority of both classes. We still do not rescale, so each threshold here is nominal and up to $\times 2.67$ optimistic.

Linewidth convention. The statistic is the single-channel flux density after drift correction, so the quoted $\text{EIRP}_{5\sigma}$ applies to any transmitter narrower than the native channel and is flat in the assumed linewidth. Two sub-channel qualifications attach. Flatness is exact only for a centred tone, a boundary-straddling one splitting its power between two channels at worst by a factor of two. And the channel-to-channel transition follows the correlator’s passband shape, which the frozen products do not retain. Thresholds therefore carry a sub-channel-position systematic of tens of per cent, and EIRP values are quoted to two significant figures.

Transmitters wider than one channel. The baseline filter (§4.2) is flat for features narrower than roughly half its effective width W and falls to zero above W , so the survey’s sensitivity is to channel-confined morphologies and W is the linewidth ceiling: 65–74 channels over the fine class and 30–59 over the coarse one, that is ≈ 32 MHz

at 488 kHz and ≈ 0.9 GHz at 15.62 MHz. Within that ceiling, a top-hat transmitter spanning N_{ch} native channels has per-channel flux the total divided by N_{ch} at unchanged noise, so to first order

$$\text{EIRP}_{5\sigma}(N_{\text{ch}}) \simeq N_{\text{ch}} \times \text{EIRP}_{5\sigma}(1), \quad N_{\text{ch}} \lesssim W/2, \quad (\text{A1})$$

the rescaling the main text applies to signals broader than one channel.

Worked example: from flux to EIRP. Take UV Ceti (G 272-61B) Band 3, the deepest of its 4 windows in that band (execution block A002_X877e41_X14c1, lower edge 91.565 GHz). Its per-channel rms is 0.241 mJy, so $S_{\text{min}} = 5 \times 0.241 = 1.21$ mJy in one 15.625-MHz native channel, or $1.21 \times 10^{-29} \text{ W m}^{-2} \text{ Hz}^{-1}$ at 2.67 pc, and $\text{EIRP}_{5\sigma} = 4\pi d^2 S_{\text{min}} \Delta\nu = 1.6 \times 10^{13} \text{ W}$.

Intra-integration smearing. The frequency excursion at the maximum searched drift rate across one integration, $\dot{\nu}_{\text{max}} t_{\text{int}}$, is at most 1.10 channels across the 443 windows (median 0.001, 90th percentile 0.06). A linear drift smeared by Δ channels within an integration keeps a fraction $\eta_{\text{smear}} = \text{sinc}(\pi\Delta/2)$ of the peak-channel amplitude, so the effective threshold is $\text{EIRP}_{5\sigma}/\eta_{\text{smear}}$. The 9 windows with $\eta_{\text{smear}} < 0.99$, as star, band, (η_{smear} , threshold multiplier), are HIP 10679 Band 6 (0.574, $\times 1.74$), β Pic Band 6 (0.574, $\times 1.74$), HD 172555 Band 6 (0.574, $\times 1.74$), HIP 10679 Band 6 (0.606, $\times 1.65$), HD 172555 Band 6 (0.606, $\times 1.65$), η Crv Band 8 (0.865, $\times 1.16$), η Crv Band 7 (0.932, $\times 1.07$), HD 48370 Band 6 (0.979, $\times 1.02$) and CP-72 2713 Band 6 (0.979, $\times 1.02$). The remaining 434 agree with their nominal thresholds to better than 1 per cent. Compounding the worst smearing penalty with the worst Hanning factor gives $\times 4.6$ on the nominal threshold for those five windows, and no released column carries that product. The $\text{sinc}(\pi\Delta/2)$ form is the conservative reading: a pure occupancy argument would retain $\min(1, 1/\Delta)$, giving 1.0 rather than 0.64 at $\Delta = 1$.

B. INJECTION-RECOVERY SENSITIVITY VALIDATION

To test whether a 5σ threshold behaves as idealised Gaussian noise predicts, we inject synthetic tones of known amplitude into real visibilities and measure what the unmodified pipeline returns. The injection uses the extraction code’s own mechanism, as far upstream of the statistic as the retained products allow. Six configurations span three bands and both classes: three coarse Band 7 spectral windows of AT Mic, one coarse window each in Bands 3 and 6 of AU Mic, and AU Mic’s 488-kHz flagband window with its 49-trial drift grid. Each receives a complex tone at the star position at amplitudes 4–10 $\times$ the statistic’s own noise. The grid covers drift fractions $f_{\text{drift}} = 0$ to 1.0 of the ceiling, on- and half-grid rates, and channel-centred and boundary placements, 200 trials per configuration and 1200 in all. The recovery criterion was frozen in advance: $T \geq 5$, peak channel within 3 of the injection, peak drift within one grid step.

The coarse-window variant recovered 0 of 500 injections from 4σ to 10σ under that criterion, which is the one withdrawn result here: the criterion produced the artefact and the pipeline suppresses nothing. Its drift-matching clause cannot be satisfied on coarse windows, where every in-grid drift is sub-channel over the track, so

the reported peak drift is arbitrary. Scored on detection alone the same windows recover essentially everything, an end-to-end injection through the released pipeline returning T_* of 20.5, 65.3 and 217.7 at 1, 3 and 10 times the per-integration noise against an ideal stacking gain of $\sqrt{639} = 25$. Class B limits therefore bound carriers of any dwell fraction, with the completeness of §4.5, and what those windows lack is drift discrimination. The fine window fails the same criterion the same way.

The second result is the completeness curve. On the fine window, drifting carriers ($f_{\text{drift}} \geq 0.5$, pooled over grid phase and sub-channel placement) are recovered 17, 42, 58, 75 and 83 per cent at 4, 5, 6, 8 and 10σ , saturating by twice threshold power (Fig. 7). The machinery alone, with the tone injected after the baseline step, is amplitude-faithful to within $-9/+14$ per cent for channel-centred tones. Boundary-straddling placement costs 0.6–0.8 in the statistic and recovers 0 per cent at 5σ against 83 per cent for centred ones, crossing 50 per cent near $1.4\times$ threshold amplitude. Survey completeness uses the placement-pooled curve, conservative for the boundary-straddling case: $\text{EIRP}_{50} = 1.10 \text{EIRP}_{5\sigma}$ and $\text{EIRP}_{90} > 2.0 \text{EIRP}_{5\sigma}$, a bound only.

This is a demonstration that the pipeline recovers a carrier in a representative configuration. It is not a survey completeness function. The curve is measured on one window; the stratified campaign below measures how far it travels to the other 126 fine-channel windows. Of the 126 Class A windows, 58 (46 per cent) differ from that configuration on at most one axis, the mismatch being dominated by band (63 windows) and drift-trial count (63) rather than channel width (17), integration time (20) or beam offset (12). Those are the axes the campaign below spans. Not spanned at all are other target environments and edge-phased drift grids.

The stratified campaign. Those axes have since been measured. A drifting carrier was injected into the retained per-integration spectra of 28 further windows, one per star, spanning Bands 3–10, 15.3–977 kHz channels, 45–485 integrations, 7–1186 drift trials and both the 12 m array and the ACA, pushed through the pipeline’s own baseline removal and de-drift stack on the search’s drift grid: nine amplitudes, 4032 trials. Recovery is a threshold crossing at the injected frequency, not the global maximum of the grid: requiring an injection to beat a window’s brightest cell measures its worst artefact, the conflation behind the withdrawn result above.

Pooled recovery is 34, 61, 89 and 99 per cent at 4, 5, 6 and 8σ , $P_{50} = 4.6\sigma$, $P_{90} = 6.1\sigma$ (Fig. 7). The spread across configurations is what the transfer bracket needed: P_{50} runs $0.64\text{--}1.13\times$ its median and P_{90} $0.48\text{--}1.35\times$ its own, so the assumed $0.5\text{--}2\times$ bracket is conservative. Band and array matter least, P_{50} varying only $4.4\text{--}4.8\sigma$ across bands.

How far the grid reaches. 28 configurations are a sparse grid for 126 windows, so we state plainly which windows the bracket interpolates and which it extrapolates. The grid spans Bands 3, 6, 7, 8 and 10, channel widths 15.3–977 kHz, 45–485 integrations and both arrays. Every Class A window lies in a band the grid samples; 108 of the 126 (86 per cent) lie inside it on all three axes at once. The 18 that do not are extrapolations, 17 of them in integration count alone (shorter or longer tracks than

TABLE 10

SELECTION FUNCTION OF THE SEARCHED SAMPLE AGAINST A VOLUME-LIMITED REFERENCE CENSUS: GAIA DR3 SOURCES WITH $\varpi > 0$, $\sigma_\varpi/\varpi \leq 0.1$ AND $1/\varpi \leq 40$ pc (17566 OBJECTS, OF WHICH 8594 CARRY A **TEFF_GSPHOTO** ESTIMATE, 49 PER CENT), AGAINST THE 88 STARS WITH AT LEAST ONE SEARCHED WINDOW. TEMPERATURES ARE AVAILABLE FOR 52 OF THE 88 (ALL 88 MATCHED TO THE 168-ENTRY ALMA-COVERED CENSUS, 58 BY EXACT NAME; 52 OF THOSE MATCHES CARRY A TEMPERATURE), SO THE TEMPERATURE PANEL IS NORMALISED OVER CLASSIFIED OBJECTS ONLY. “RATIO” IS THE SAMPLE COLUMN FRACTION OVER THE CENSUS COLUMN FRACTION; > 1 IS OVER-REPRESENTATION.

Property	Census	Searched	Ratio
Distance (all 88 searched stars; 17566 census objects)			
$d \leq 5$ pc	48 (0.3%)	12 (13.6%)	49.9
5–10 pc	240 (1.4%)	13 (14.8%)	10.8
10–20 pc	2069 (11.8%)	18 (20.5%)	1.7
20–30 pc	5391 (30.7%)	18 (20.5%)	0.7
30–40 pc	9818 (55.9%)	27 (30.7%)	0.5
T_{eff} class ^a (52 searched; 8594 census)			
A (≥ 7500 K)	24 (0.3%)	1 (1.9%)	6.9
F (6000–7500 K)	402 (4.7%)	9 (17.3%)	3.7
G (5300–6000 K)	860 (10.0%)	16 (30.8%)	3.1
K (3900–5300 K)	1400 (16.3%)	5 (9.6%)	0.6
M (< 3900 K)	5908 (68.7%)	21 (40.4%)	0.6
Other axes			
Known planet hosts ^b	20 (11.9%)	18 (20.5%)	1.7
Searched stars / systems	–	88 / 81	–

^aBins are temperature bins, not MK types: A ≥ 7500 , F 6000–7500, G 5300–6000, K 3900–5300, M < 3900 K. Gaia **teff_gspphoto** is unavailable for about half the reference census, preferentially for the coolest and faintest objects, so the census M fraction is a lower limit and the earlier-type ratios are correspondingly upper limits. White dwarfs are not separable in this parameterisation and fall wherever their (unreliable) photometric temperature places them.

^bThe Gaia reference census carries no planet information; the census column here is instead the 168-entry ALMA-covered census, of which 20 entries are known planet hosts. The 88 searched stars include 18 hosts of 41 known planets.

any calibrated configuration) and 1 in channel width. The transfer bracket should be read as measured for the 86 per cent and assumed for the rest. The five deepest Class A thresholds in the survey are all interpolated, so the headline sensitivity does not rest on an extrapolation.

The spectral response, measured. The $\times 2.29$ correction of §4 was analytic because the injected tones were unsmoothed. Injecting a tone at fractional position ϕ , split between neighbouring channels as the delivered channelisation requires and convolved with the 0.25/0.5/0.25 kernel, and comparing its peak with a channel-centred unsmoothed tone of equal total flux, returns $\times 2.02$ at $\phi = 0$ and $\times 2.69$ at the channel edge, median $\times 2.31$ over 400 trials, every point within 1.1 per cent of the analytic value. Both campaigns ship trial-level records.

C. PER-TARGET RESULTS

The per-window results ship as the comma-separated catalogue of the Data Availability statement. Its $\text{EIRP}_{5\sigma}$ column uses the definition of §4, on the same threshold and primary-beam correction as the continuum column, so a row’s two limits are comparable; its per-channel flux-density threshold uses the carrier lane’s per-channel noise rather than the tabulated **rms_mJy**, the two differing by $5 \times G(r)$. Bands 6 and 7 supply 84 per cent of the windows.

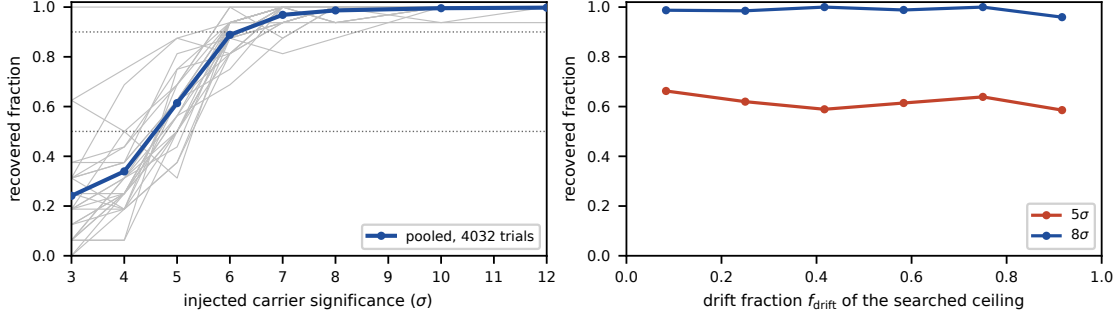


FIG. 7.— Stratified injection recovery. Grey, each of the 28 configurations separately; blue, the pooled curve over 4032 trials. Dotted lines mark the 50 and 90 per cent levels. The spread between grey curves is the completeness transfer that was previously assumed rather than measured. Right, recovery against drift fraction of the searched ceiling at 5σ and 8σ , the axis the single-configuration measurement used to carry alone.

TABLE 11

PER-BAND SURVEY CHARACTERISATION OF THE 443 SEARCHED WINDOWS. “Sys.” COUNTS UNIQUE STELLAR SYSTEMS SEARCHED IN THE BAND, SO A SYSTEM OBSERVED IN TWO BANDS IS COUNTED IN BOTH ROWS (81 SYSTEMS, 88 STARS IN TOTAL); “EBs” THE CONTRIBUTING EXECUTION BLOCKS, ATTRIBUTED BY THE PROVENANCE OF THE DE-DUPLICATED WINDOWS; $\Delta\nu_{\text{ch}}$ THE NATIVE CHANNEL WIDTH IN MHz (MEDIAN, RANGE IN BRACKETS); “UNION” THE UNIQUE SKY FREQUENCY SEARCHED IN THE BAND AFTER MERGING OVERLAPPING WINDOWS; $\text{EIRP}_{5\sigma}$ THE MEDIAN NOMINAL THRESHOLD ACROSS THE BAND’S WINDOWS; $\dot{\nu}_{\text{ceil}}$ THE RANGE OF DRIFT-RATE CEILINGS IN Hz s^{-1} PER GHz OF CARRIER FREQUENCY. THE “ALL” ROW IS COMPUTED OVER THE WHOLE SAMPLE, NOT SUMMED DOWN THE COLUMNS: EBs AND SYSTEMS RECUR ACROSS BANDS AND THE FREQUENCY UNIONS ARE DISJOINT BY CONSTRUCTION.

Band	Sys.	EBs	Win.	$\Delta\nu_{\text{ch}}$ (MHz)	Union (GHz)	Med. $\text{EIRP}_{5\sigma}$ (W)	$\dot{\nu}_{\text{ceil}}$
3	8	9	40	15.63 (0.244–15.63)	20.6	1.1×10^{14}	12.0–13.3
4	1	1	4	15.63	6.9	2.6×10^{13}	12.0
5	1	1	4	15.63	6.9	4.4×10^{13}	12.0
6	54	54	216	15.63 (0.015–31.25)	29.4	1.3×10^{15}	12.0–13.3
7	33	34	155	15.63 (0.061–15.63)	22.6	2.9×10^{15}	12.0–13.3
8	3	3	12	15.63 (0.061–15.63)	6.6	6.1×10^{16}	12.0
9	1	1	4	15.63	6.9	4.3×10^{15}	12.0
10	1	1	8	0.977	13.8	2.7×10^{16}	12.0
All	81	104	443	15.63 (0.015–31.25)	113.9	1.5×10^{15}	12.0–13.3

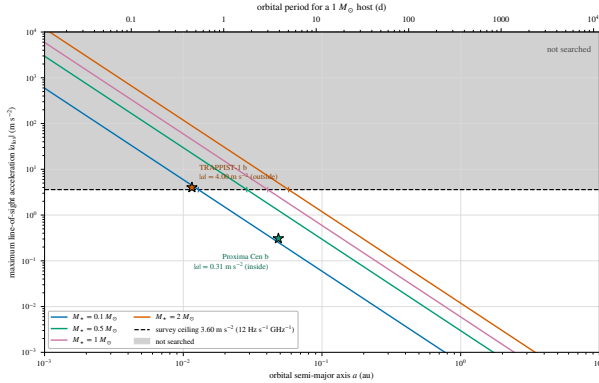


FIG. 8.— The drift-rate ceiling as a physical selection function. Maximum star–planet relative acceleration, edge-on and maximised over orbital phase, as the carrier drift rate it induces (Hz s^{-1} per GHz of carrier frequency; equivalently $\dot{\nu}/\nu = \dot{\nu}/c$), against orbital period, for host-star masses 0.09 – $1.8 M_{\odot}$ (lines) and the known planets of the planet-hosting targets (points). The shaded region lies above the drift-search ceiling, outside the searched grid: $a < 0.041$ au ($P < 3.0$ d) for a $1 M_{\odot}$ host. Of 41 known planets only TRAPPIST-1 b reaches it (§6).

D. THE VELOCITY-SPACE LINE MASK

The one-channel line check failed the β Pictoris windows (§5.3): a native channel spans a fraction of a km s^{-1} , while circumstellar kinematics shift real emission by many channels. The replacement mask takes its tolerance from the kinematic budget of circumstellar gas: Keplerian motion where these molecules survive ($\gtrsim 10$ au), $\lesssim 13 \text{ km s}^{-1}$; cold-gas linewidths $\lesssim 1 \text{ km s}^{-1}$; cold dense-gas tracers with no bulk-wind term for the dwarfs domi-

TABLE 12

BAND-BY-BAND COST OF THE MOLECULAR-LINE EXCLUSION. UNION: UNIQUE SKY FREQUENCY SEARCHED IN THE BAND. MASKED: THE PART OF IT INSIDE A $\pm 50 \text{ km s}^{-1}$ STELLAR-FRAME TUBE AROUND A CATALOGUED TRANSITION OF ONE OF THE MASKED SPECIES. LEFT: WHAT A TECHNOSIGNATURE SEARCH CAN STILL OCCUPY. THE MASK IS EXCLUDED SEARCH SPACE AND NOT A VETO (§5.3.10); THE LOSS IS CONCENTRATED IN BAND 6, WHERE $\text{CO } 2 \rightarrow 1$ SITS.

Band	Union (GHz)	Masked (GHz)	Left (GHz)	Lost (%)
3	20.64	0.272	20.37	1.3
4	6.94	0.049	6.89	0.7
5	6.94	0.065	6.87	0.9
6	29.44	1.374	28.07	4.7
7	22.60	1.252	21.35	5.5
8	6.55	0.164	6.39	2.5
9	6.94	0.454	6.48	6.5
10	13.80	0.844	12.96	6.1
All	113.86	4.474	109.38	3.9

nating the sample; and an uncorrected barycentric term bounded by $\pm 30 \text{ km s}^{-1}$. The result is empirically conservative: the largest attributed circumstellar component (β Pic, within 0.46 km s^{-1} of systemic) sits at 0.9 per cent of it, and *zero* crossing frequencies lie 13 – 50 km s^{-1} from their nearest *masked* transition (the wider catalogues are another matter: §5.3), so a $\pm 13 \text{ km s}^{-1}$ Keplerian-only mask would have suppressed and released the same windows: the wider tolerance costs no yield and buys robustness against radial-velocity catalogue error. The alternative selection quoted in §5.3.10 is every Splatologue transition over the 45 islands with $E_u \leq 150 \text{ K}$ and $\log A_{ij} \geq -5$, 4,958 of 233 carriers; HCN, CS, SiO, CN and H_2CO are in the adopted list already, and no cover-

age figure and no disposition changes.

The repaired mask. The four defects and their repair are in §5.3.10. The rounding defect is why the released CO(1→0) offset column, referred to the frozen mask’s rounded 115.271000 GHz entry, reads 0.202 MHz (0.53 km s^{-1}) smaller than the laboratory-referred value of Table 15. A cheaper rule exists: the archive records the rest frequency each execution block was tuned to, so masking each window’s own tuned line, inside an LSR-frame tube of a few $\times 10 \text{ km s}^{-1}$, catches the β Pictoris, HD 48370 and [C I] windows automatically.

The crossing ledger, once: of the 20 windows with an on-star crossing, 4 also beat their control ring and three of those are accounted for by the mask. Two further crossing windows fall inside the mask without beating their ring (the unflagged β Pic Class A window at -0.46 km s^{-1} ; HD 285968 at $+8.7 \text{ km s}^{-1}$, already rejected by its ring, maximum 10.9 against a star peak of 6.5), and the remainder lie far outside any circumstellar tolerance (AU Mic $+379$, LHS 1140 $+115 \text{ km s}^{-1}$). The other 16 are ordinary noise crossings: in each, the window’s own control ring reaches or exceeds the stellar value (ring-maximum median 5.84 against a crossing median 5.18), the expected Gaussian-tailed population of moderate $5\text{--}5.5\sigma$ crossings with comparably strong peaks among 512 controls. But the transitions that make astrophysical false positives cheap are also natural reference frequencies for a deliberate communicator (Cocconi & Morrison 1959), so the released products keep a known-line-coincident list: every on-source 5σ crossing inside the masked intervals, with its offset from each nearby transition.

E. THE ANCILLARY CONTINUUM LANE

The continuum lane is an exploratory by-product: 57 measurements, 17 detections and 40 limits. A millimetre excess is at best an indirect and model-dependent observable of waste-heat technology. An “upper limit” here is $5\times$ the primary-beam-corrected map rms at the star’s position, and the deepest non-detection is towards TRAPPIST-1 in Band 3. Under the combined error model sixteen of the seventeen detections stay above 5σ , the exception being η Corvi Band 7 at 4.7σ .

Every detection is astrophysically unsurprising: the unresolved UV Ceti binary; six systems of disc or chromospheric emission (β Pic, AU Mic, AT Mic and τ Ceti among them); six stars at photospheric flux; and two late-M dwarfs of documented magnetic activity, LP 349–25 and the auroral emitter LSR J1835+3259 (Hallinan et al. 2015). Closure phase discriminates structure for none of them at these flux densities.

The anomaly screen. Each star’s flux is tested by a robust median/MAD statistic against its own photospheric expectation and against same- T_{eff} stars in coarse bins of at least five. The error model, fixed before any flag, adds the map noise, a calibration term of 5 per cent in Bands 3–6 and 10 per cent in Bands 7–8, and a model term propagated from the Gaia temperature and radius errors. It has *no intrinsic-variability term*. That is adequate for a photosphere and wrong for an active M dwarf, whose millimetre flux can vary by factors of several, so the screen is uninformative for exactly the stars a radio-stellar reader cares about. It flags for inspection and

tests no per-star hypothesis. One outlier results, χ^1 Ori at Band 3, consistent with a spectroscopic G0V binary’s unresolved companion or a chromosphere.

F. FALSE-ALARM ACCOUNTING UNDER THE SYMMETRIC CRITERION

The per-window rate is $1/513$ and the survey expectation $443/513 = 0.86$ flags. The floor falls only if the control ensemble grows, which the primary beam bounds. Windows are not independent trials either, grouping into 104 execution blocks that share weather, calibration and correlator setup, and that clustering is measured here rather than assumed. Drawing one control index per execution block and applying it to all of that block’s windows gives the clustered distribution in closed form: mean 0.87 rank-first windows, and under the 5σ gate 0.26 stage-1 outliers with $P(\geq 1) = 0.23$ for the single unattributed one. Clustering changes little because the *position* of the control maximum is not shared across a block’s windows.

The trials budget, in one set of units, is Table 13. A window holds $177\text{--}6866208$ channel $\times$ drift cells (median 468), the survey 4.80×10^7 in total, so a one-sided Gaussian 5σ tail predicts 13.8 chance cells across the survey, an expectation in cells and not in windows. Against that, 229 on-star cells reach 5σ in 20 windows, 174 of them in the four stage-1 outliers; the remaining 55 stand against 13.8 expected. Correlation cannot move the expected *number* above 5σ , which is Np whatever the correlation, only its clumping, so the factor decomposes into cell multiplicity within a crossing window and the count of crossing windows.

The control ensembles calibrate this without a Gaussian assumption: the same arithmetic predicts 143 of the 443 windows carrying a control above 5σ and 134 do, agreeing to 6 per cent, with per-window control maxima tracking the iid-Gaussian prediction at a median ratio 0.995 (5th–95th percentile 0.91–1.11). The residual bookkeeping is that n_{cells} counts grid cells and not independent trials: Hanning smoothing correlates neighbouring channels at $\rho_1 = 2/3$, $\rho_2 = 1/6$, so $1+2\rho_1+2\rho_2 = 2.67$ grid channels carry one channel’s information, and the drift grid is oversampled two to one, giving a combined over-count $2.67 \times 2 = 5.3$. Dividing n_{cells} by it lowers the predicted maxima from 5.74 to 5.45 in the fine class and from 4.47 to 4.10 in the coarse one, against 5.77 and 4.42 observed: a residual $+6$ and $+8$ per cent in the two classes independently. Non-Gaussianity of that size is why the operative statement is the empirical rank, not the Gaussian budget.

The same correction disposes of most of the on-star cell excess. Of the factor 4.0 above, the part contributed within a crossing window is 3.4 cells observed against 1.2 expected under independence, a factor 2.8, which is what an over-count of 2.67 on the frequency axis predicts: one real threshold excursion is counted about three times. What that leaves is the window-level factor, 16 windows carrying at least one unattributed on-star crossing against 11.5 expected, a factor 1.4 whose one-sided Poisson probability is 0.12. Once the four dispositioned windows are set aside, the residual excess is consistent with chance at that level. Table 13 collects the decomposition, so that the accounting can be audited at a glance

TABLE 13

THE ON-STAR CELL EXCESS, DECOMPOSED. TOP: WHAT IS OBSERVED AGAINST AN IID-GAUSSIAN BUDGET. MIDDLE: THE TWO MULTIPLICATIVE PARTS OF THAT FACTOR, THE FIRST PREDICTED BY THE HANNING CHANNEL CORRELATION AND THE SECOND A WINDOW COUNT CONSISTENT WITH CHANCE. BOTTOM: THE SAME CORRELATION CORRECTION APPLIED TO THE CONTROL MAXIMA, WHICH ARE MEASURED IN THE SAME UNITS AND ARE WHAT CALIBRATES THE BUDGET EMPIRICALLY.

Quantity	Value	Factor
On-star cells $\geq 5\sigma$	229	
in the four stage-1 windows	174	
remaining, unattributed	55	
iid-Gaussian expectation (cells)	13.8	4.0
Cells per crossing window (obs./exp.)	3.4/1.2	2.8
predicted by Hanning correlation	2.67	
Crossing windows (obs./exp.)	16/11.5	1.4
one-sided Poisson p	0.12	
Frequency-axis over-count $1 + 2\rho_1 + 2\rho_2$	2.67	
Drift-grid oversampling	2	
Combined cell over-count		5.3
Control max., fine: pred. $\rightarrow$ corrected	5.74 $\rightarrow$ 5.45	
observed	5.77	+6%
Control max., coarse: pred. $\rightarrow$ corrected	4.47 $\rightarrow$ 4.10	
observed	4.42	+8%

rather than followed through four paragraphs.

It is not zero, runs one way, and carries the same sign at the controls, so we report it as an open systematic. The crossing list is dominated by CO(2 $\rightarrow$ 1) at 230.5 GHz toward disc hosts, and that a residual excess of on-star chance crossings at the tens-of-per-cent level cannot be excluded. The operative false-alarm statement remains the rank against the control ensemble.

This fixes what a single window can ever mean. The rank floor is $1/(N_{\text{ctrl}} + 1) = 1.95 \times 10^{-3}$ against a survey-wide Bonferroni scale of 1.1×10^{-4} , a factor 16, and correlation widens it: with $N_{\text{eff}} \simeq 30\text{--}43$ independent spatial trials the ensemble resolves only 0.023–0.032. Drawing more controls would not help, the primary beam bounding the annulus. This is the design’s central limitation and is why authentication rests on recurrence and astrophysical attribution rather than on rank.

The control geometry uses a 12-m θ_{PB} for every window, so the 141 ACA 7-m windows (32 per cent of the survey, array read from the archive’s antenna list rather than assumed) carry a known geometric error. It runs in the conservative direction: on the true 7-m beam those annuli span 0.08–0.46 of θ_{PB} at a median area-weighted gain of 0.75 against 0.47 for the 12-m windows, so exchangeability in flux density improves there. Of the four stage-1 outliers only HD 48370 Band 6 is ACA.

Dropping the ACA windows tests what that geometry costs, and Table 14 gives the two strata side by side. The 12 m stratum holds 3 of the 4 stage-1 outliers, the one lost being HD 48370’s foreground CO; the deepest trigger and the deepest P_{eff} are both 12 m windows and do not move, so no survey-level statement changes. The ACA stratum alone departs mildly ($D = 0.13$, $p = 0.018$), opposite in direction to what a mis-scaled annulus gain would produce. Re-extracting the 141 ACA windows on the correct 7 m beam would remove the error rather than bound it; it needs the raw visibilities and a full re-search, which is beyond this cycle, and the table is what we offer in its place. Excluding the 15 β Pictoris windows leaves one unattributed crossing among 416, against 0.81 expected ($P(\geq 1) = 56$ per cent).

TABLE 14

ARRAY-SPLIT ROBUSTNESS. THE CONTROL GEOMETRY USES A 12-M θ_{PB} FOR EVERY WINDOW, SO THE ACA 7-M STRATUM CARRIES A KNOWN GEOMETRIC ERROR. EXPECTED RANK-FIRST COUNTS ARE $N_{\text{win}}/(512 + 1)$ UNDER EXCHANGEABILITY, AND ARE THEREFORE LOWER BOUNDS: THE MEASURED TAIL RATE IS A FACTOR 1.4 HIGHER (§5.3.4), WHICH BRINGS THE 12 M STRATUM’S 3 OBSERVED WITHIN ONE OF EXPECTATION. NO DISPOSITION AND NO HEADLINE THRESHOLD DEPENDS ON THE ACA WINDOWS.

	12 m	ACA 7 m
Windows	302	141
Execution blocks	66	38
Stars; systems	62; 55	35; 35
Threshold crossings	8	12
Rank-first windows	3	2
expected	0.59	0.27
Stage-1 outliers	3	1
Stellar-rank $D(p)$	0.047 (0.50)	0.13 (0.018)

F.1. The window-level and survey-level rank calibration

The drift grid and the thousands of channels apply identically at the star and at every control, so the per-window trials factor is absorbed empirically: under exchangeability one designated position of 513 ranks first with probability exactly $1/513$ whatever the noise distribution, and a heavy tail cannot by itself produce an excess of rank-first windows. The survey expectation is 0.86 and 5 are observed; the execution-block clustered permutation returns $P(\geq 5) = 1.9 \times 10^{-3}$, and the excess is astrophysical, celestial line emission sitting at the stellar position in 3 of the 5 (Table 6) and leaving 2 against 0.86 expected. A prespecified Monte Carlo that promotes one control per window to pseudo-star and runs it through the operative gates expects 0.20 flagged pseudo-target windows, agreeing with the rank budget and the Bonferroni scale.

Ranking against 512 controls cannot resolve a probability below $1/513$, so a local scramble test was built to carry the four stage-1 windows past that floor. It scrambles each integration independently in channel and re-runs the identical search with the same noise map, weights and drift grid, which destroys anything coherent along a drift track while preserving each integration’s own spectral autocorrelation and trials factor; its validation criteria and failure mode are in Appendix G. Three of the four windows pass validation. The two β Pictoris windows behave as positive controls, no draw of 10^5 reaching T_* in either ($p < 1 \times 10^{-5}$ and $p < 5 \times 10^{-5}$), which is what a working statistic should return where a genuine localised signal is present. For CP–72 2713 the null resolves the value itself, 157 draws of 10^5 equalling or exceeding T_* , and an extreme-value fit to that window’s own control maxima agrees to a factor 1.3; both are used with that window below. HD 48370 Band 6 fails validation outright, 73 per cent of its real controls exceeding the maximum of 10^5 null draws, because the field carries extended emission and imaging residuals that are coherent across integrations and that the scramble destroys by design, so we quote no scrambled-noise probability there. That failure is a direct consequence of retaining full dynamic spectra for only 8 of the 512 control positions, and no number quoted here rests on a parametric extrapolation of a null tail.

Exchangeability is the statistic’s central assumption, and it can fail through spatially varying noise, sidelobes,

correlated pixels or calibration residuals at the phase centre. For a debris-disc archive the dominant violation is none of those but resolved astrophysical emission filling the annulus, seen here in HD 48370’s CO ring and in β Pictoris’ recurring Band 6 window, where the ring outshines the star by $1.5\times$; that suppresses a star-exceeds-ring outcome instead of manufacturing one, so it costs sensitivity and nothing else; the repeat block of the *flagged* Band 6 window measures that cost (§5.3.7). The primary beam is settled by construction, since the statistic is formed on *uncorrected* amplitudes whose thermal noise does not depend on position and $\sigma(t, \nu)$ is one scale shared by the star and every control (§4.2), so the 513 positions are *approximately* exchangeable under a *restricted* noise-only null. Both qualifiers carry weight, because that argument covers the thermal noise and the beam and does not cover residual calibration or continuum systematics, which can occur preferentially where the pointed science target sits.

Four measurements bound what is left, and all four are null. Ranking each control against the other 512-1 by the rule used for the star gives 220 672 pseudo-star ranks, uniform to the resolution the ensemble can express ($p = 0.37$) in every band and at both channelisations; that geometry cannot by itself detect a centre-versus-annulus asymmetry, so the test that can is the two-sample comparison of the 443 real stellar ranks against the pooled pseudo-ranks, and it returns $p = 0.38$. On that comparison the stellar position behaves like one more position in its own annulus, on these data; out of sample it does not (§5.3.1). The Spearman correlation of control statistic against reconstructed radius has median 0.009 over the 307 crossing-free windows, bounding any systematic radial suppression of the controls below 11 per cent in the mean. And the star’s own residual variance is measured directly against the controls’ in §5.3.5.

One stratum departs, and we report it separately. The fine-channel windows fail the $U(0,1)$ rank test at $p = 0.015$ ($n = 126$), because four of them contain an on-star peak above their own control maximum against 0.22 expected, and those four are precisely the flagged windows of Table 6. That fits the astrophysical line emission identified below, with no misbehaving ensemble, but the departure is genuine, so fine-window rank probabilities here are slightly optimistic; excluding the 20 fine windows with any $\geq 5\sigma$ stellar crossing restores uniformity ($p = 0.39$). Two stratifications the frozen products cannot support, per-control noise records and gain, are bounded only through these empirical rank distributions. Figure 9 shows the by-stratum behaviour: median per-window control maxima are flat across bands and the fine class sits higher than the coarse, 5.77 against 4.42, as its far larger channel $\times$ drift trial count predicts.

A fully symmetric region-max variant, which maximises identically at the star and at every control, can be run on only seven locally reprocessed windows because the frozen release does not carry the products it needs. In all seven the star is exchangeable with its ring (Appendix G). The single-position calibration above remains the operative release-wide criterion, and the two are never interchanged.

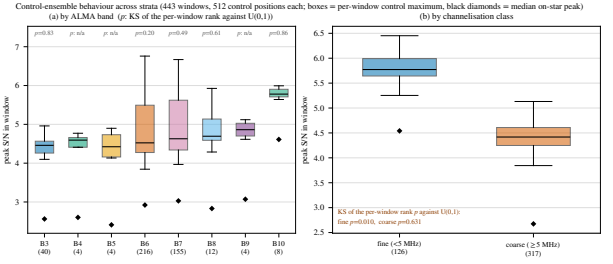


FIG. 9.— Exchangeability of the 512-position control ensemble by stratum. Boxes: per-window control maximum; diamonds: median on-star peak; p : the Kolmogorov–Smirnov test of the text. The ensemble tracks the on-star statistic in every band and both channelisation classes. The one departure is the drift-search class ($p = 0.015$), driven by the four flagged windows of Table 6.

G. ANALYSIS PROVENANCE AND ROBUSTNESS CHECKS

G.1. The R_σ residual-scale test

For each of the four stage-1 windows the pipeline’s own baseline-subtracted per-integration residuals at the star and at the 8 retained control positions are standardised by its noise map $\sigma(t, \nu)$ and reduced to a robust scale over the (integration, channel) plane, the crossing channels excluded at every position alike: $R_\sigma = \sigma_{*,\text{resid}}/\text{median}(\sigma_{\text{ctrl},\text{resid}})$ is 1.000 (CP–72 2713), 0.995 (HD 48370), 1.001 (β Pic B3) and 1.001 (β Pic B6), the one star measurably *quieter* than its ring being HD 48370, whose ring is full of CO. Eight draws set the position-to-position scatter, hence a bound at the per cent level and not a tenth of one, and their own spread is 0.6 per cent; retaining all 512 control spectra (§6.4) would fix that and the test’s blindness to a channel-confined artefact together.

G.2. The fully symmetric region-max check

A fully symmetric region-max variant applies identical source-region, drift-rate and frequency maximisation at the star and at every control. It needs products the frozen release does not carry, so its domain is seven locally reprocessed windows. All seven give $T_* = 3.7\text{--}4.9$ against control maxima of $6.7\text{--}7.7$, $p = 0.81\text{--}1.00$, so the star is exchangeable with its ring wherever the test runs at full symmetry. The single-position calibration of §5.3 remains the operative release-wide criterion and the two are never interchanged (Appendix F).

G.3. The β Pictoris frame audit

Table 15 carries the frame chain for the line-attributed windows. It recovers the whole check list untuned: frequencies to the frame chain’s stated tolerance, velocity concordance across epochs and transitions, localisation against the control ensemble, and drift class zero to within the grid.

Two further properties fit CO and not a carrier. The stage-1 Band 6 window carries 114 formal $\geq 5\sigma$ crossings where a channel-confined carrier admits one. The retained products do not record whether those channels are contiguous, so the count sets a floor of 2.26 km s^{-1} on the feature’s extent without measuring it. The peak flux densities, 54 and 78 mJy per channel, are the right order for this belt’s spatially integrated CO. They are per-beam peaks on emission resolved in both beams, and Matrà et al. (2017) find the gas subthermally excited, so

TABLE 15

LINE-ATTRIBUTION AUDIT THROUGH THE FRAME CHAIN OF §4.3: THE THREE LINE-ATTRIBUTED WINDOWS, AND THE CLASS A β PICTORIS BAND 6 WINDOW THAT IS NO STAGE-1 OUTLIER YET SHOWS THE SAME CO COMPONENT. TOPOCENTRIC OFFSETS ARE OBSERVED MINUS CATALOGUED LABORATORY REST FREQUENCY (PICKETT ET AL. 1998), SO A NEGATIVE OFFSET MEANS THE OBSERVED FREQUENCY LIES BELOW REST. THE BARYCENTRIC CORRECTION IS THE EARTH-EPHEMERIS TERM AT THE EXECUTION-BLOCK MID-TIME. THE AU MIC SYSTEMIC VELOCITY IS CATALOGUED (FOUQUÉ ET AL. 2018). FOR β PICTORIS WE ADOPT $v_{\text{sys}} = +20.0 \pm 0.7 \text{ km s}^{-1}$ (MATRÀ ET AL. 2017) RATHER THAN THE *Gaia* DR3 VALUE OF $+16.84 \text{ km s}^{-1}$ (GAIA COLLABORATION ET AL. 2022), BECAUSE THE STAR IS AN A6V ROTATOR INSIDE THE HOT-STAR TEMPLATE REGIME IN WHICH DR3 RADIAL VELOCITIES CARRY A KNOWN SYSTEMATIC (BLOMME ET AL. 2023).

Flag	UTC start	f_{obs} (GHz)	Transition	Topo. offset (MHz; km s^{-1})	Bary. corr. (km s^{-1})	v_{sys} (km s^{-1})	Stellar-frame offset (km s^{-1})
β Pic B3	2022-03-03 21:49	115.260644	CO(1 $\rightarrow$ 0)	-10.56; -27.46	-7.90	+20.0	+0.44
β Pic B6	2019-03-12 00:14	230.516128	CO(2 $\rightarrow$ 1)	-21.87; -28.44	-8.15	+20.0	-0.29
AU Mic B6	2014-08-18 04:01	230.825230	CO(2 $\rightarrow$ 1)	+287.23; +373.51	-10.02	-4.71	+378.82
β Pic B6 [†]	2013-10-06 06:23	230.528134	CO(2 $\rightarrow$ 1)	-9.87; -12.83	+7.63	+20.0	-0.46

[†]Class A Band 6 window of a different execution block, without a stage-1 flag (Appendix C), shown because it independently shows the same CO component.

beam dilution and excitation both depress their ratio below optically thin LTE.

Stellar emission is a competing explanation for any stage-1 outlier towards an active young star, and the search suppresses it twice. Gyrosynchrotron and free-free emission from chromospheres and coronae are broadband, while the block-median baseline of §4.2 removes everything broader than its 65-channel passband, $\sim 32 \text{ MHz}$ at the fine class, a fractional bandwidth of $\sim 10^{-4}$; MacGregor et al. (2020)’s AU Mic flares are coherent across the full 8 GHz. Flares are also time-confined, so the inverse-variance stack over all integrations dilutes them. What survives is the multiplicative route of §4.2, a flare continuum times a spectral calibration residual, bounded there and measured for CP-72 2713 in §5.3. The search is blind to the time axis by construction, and the retained dynamic spectra are the only handle on it.

HD 48370 in Band 8. The carbon tuning says the same thing by omission. All twelve Band 8 windows here belong to η Crv, HD 48370 and HD 61005 and are [C I] $^3P_1-^3P_0$ (492.160651 GHz) tunings. HD 48370’s carries a 5.05σ on-star crossing against a ring maximum of 6.23, so it is not one of the four stage-1 outliers. Its crossing channel sits -129 km s^{-1} from [C I], so it is not [C I] emission from the disc, the cloud or the star, and its own ring dispositions it as an ordinary noise crossing. What the window does show is no line at the star in this disc’s carbon tuning, which is what Cataldi et al. (2023) report. The crossing peak in the flagged window itself is 0.82 Jy

in one 122-kHz channel with the ring at the same level.

G.4. Entries excluded from the released catalogue

The four classes of excluded entry. (i) Four target/bands (Barnard’s Star [B7], Wolf 359 [B7], SCR J1845-6357, Wolf 358) have no carrier result from a process-level failure, each keeping a valid continuum non-detection and no EIRP threshold. (ii) Giclas 9-38A and B [B3] are excluded entirely, the pair’s only matched MOUS pointing ~ 52 arcmin from either star, $35\times$ its own primary-beam FWHM. (iii) Six windows across five stars (51 Eri, HD 10647, HD 139664, TWA 7, HD 92945 twice) fail the noise test of §5.2, all 15.62-MHz coarse windows whose reported rms is inconsistent with their integration by three to four orders of magnitude and sits 388-669 below the shallowest sibling of the same block and channel width, a normalisation signature. (iv) The entry “ γ Lupi” resolves to HD 139664 at 17.40 pc, not the naked-eye star; all such entries are relabelled and no measured value changes.

12 windows toward AU Mic in Band 9 and HD 61005 in Band 10, 675-873 GHz at 4.1×10^{15} - $3.0 \times 10^{16} \text{ W}$, are included; none is a detection or a spatial outlier, the largest reaching $T_{\star} = 4.91$. The quality audit is clean across all 443 windows bar four χ^1 Ori Band 3 windows flagged as elevated (0.9 - $1.1\times$ the Band 3 median, weakening their thresholds by $\lesssim 20$ per cent in amplitude).

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